DIABETIC RETINOPATHY SEVERITY CLASSIFICATION USING VISION TRANSFORMER

A MAIN PROJECT REPORT

Submitted by

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in Partial fulfillment of the requirements for
the award of the degree

of

MASTER OF COMPUTER APPLICATIONS

Under the Guidance of

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(Assistant Professor, Department of Computer Applications)

DEPARTMENT OF COMPUTER APPLICATIONS

MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR

(Affiliated to APJ Abdul Kalam Technological University)

APRIL 2026

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Program Specific Outcomes

- **PSO I**: Apply advanced technologies through innovations to enhance the efficiency of design development.
- **PSO II**: Apply the principles of computing to analyze, design and implement sustainable solutions for real world challenges.

MAPPING OF PO-PSO-SDG

1. MAPPING WITH PROGRAM OUTCOMES (POs):-

SL.NO	POs ADDRESSED	RELEVANCE TO PROJECT
1	PO1	Apply knowledge of programming, machine learning, image processing, and software engineering to develop the Automated Diabetic Retinopathy Detection System.
2	PO2	Analyze the problem of delayed or manual diagnosis of diabetic retinopathy and identify requirements for an automated medical image analysis system..
3	PO3	Design and develop a deep learning–based system capable of analyzing retinal fundus images and detecting diabetic retinopathy stages efficiently.
4	PO4	Collect and analyze retinal image datasets, evaluate model performance, and improve prediction accuracy using research and experimental methods..
5	PO6	Apply project management concepts such as planning, scheduling, and task division during the development of the automated retinal image analysis system while managing project resources efficiently.
6	PO8	Continuously learn and adopt new technologies such as deep learning, medical image analysis, and artificial intelligence to improve the performance and capabilities of the system.

LIST OF PROGRAM OUTCOMES (POs):

PO1 (Foundation Knowledge): Apply knowledge of mathematics, programming logic and coding fundamentals for solution architecture and problem solving.

PO2 (Problem Analysis): Identify, review, formulate and analyse problems for primarily focussing on customer requirements using critical thinking frameworks.

PO3 (Development of Solutions): Design, develop and investigate problems with as an innovative

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PO4 (Modern Tool Usage): Select, adapt and apply modern computational tools such as development of algorithms with an understanding of the limitations including human biases.

PO5 (Individual and Teamwork): Function and communicate effectively as an individual or a team leader in diverse and multidisciplinary groups. Use methodologies such as agile.

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PO7 (Ethics): Commit to professional ethics in managing software projects with financial aspects. Learn to use new technologies for cyber security and insulate customers from malware.

PO8 (Life-long learning): Change management skills and the ability to learn, keep up with contemporary technologies and ways of working.

2. MAPPING WITH PROGRAM SPECIFIC OUTCOMES (PSOs):

SL.NO	PSOs ADDRESSED	RELEVANCE TO PROJECT
1	PSO 1	Uses modern technologies such as deep learning, medical image processing, and web technologies to develop an automated diabetic retinopathy detection system.
2	PSO 2	Applies computing principles to analyze retinal images, design an intelligent detection model, and implement a sustainable healthcare solution for early diagnosis of eye diseases.

LIST OF PROGRAM SPECIFIC OUTCOMES (PSOs):

PSO 1: Apply advanced technologies through innovations to enhance the efficiency of design development.

PSO 2: Apply the principles of computing to analyze, design and implement sustainable solutions for real world challenges.

3. MAPPING WITH SUSTAINABLE DEVELOPMENT GOALS (SDGs):

SDG NO	SDGs ADDRESSED	RELEVANCE TO PROJECT
SDG 3	Good Health and Well-being	Supports early detection of diabetic retinopathy and helps prevent vision loss through automated diagnosis.
SDG 9	Industry, innovation and infrastructure	Utilizes advanced technologies such as artificial intelligence and deep learning to improve healthcare infrastructure and diagnostic systems..
SDG 10	Reduced Inequalities	Provides accessible screening support in areas where expert ophthalmologists or advanced medical facilities are limited.

SUSTAINABLE DEVLOPMENT GOALS (SDGs):

SDG 1 – No Poverty-End poverty in all its forms everywhere.

SDG 2 – Zero Hunger-End hunger, achieve food security and improved nutrition, and promote sustainable agriculture.

SDG 3 – Good Health and Well-Being-Ensure healthy lives and promote well-being for all at all ages.

SDG 4 – Quality Education-Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

SDG 5 – Gender Equality-Achieve gender equality and empower all women and girls.

SDG 6 – Clean Water and Sanitation-Ensure availability and sustainable management of water and sanitation for all.

SDG 7 – Affordable and Clean Energy-Ensure access to affordable, reliable, sustainable, and modern energy for all.

SDG 8 – Decent Work and Economic Growth-Promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all.

SDG 9 – Industry, Innovation, and Infrastructure-Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.

SDG 10 – Reduced Inequality-Reduce inequality within and among countries.

SDG 11 – Sustainable Cities and Communities-Make cities and human settlements inclusive, safe, resilient, and sustainable.

SDG 12 – Responsible Consumption and Production-Ensure sustainable consumption and production patterns.

SDG 13 – Climate Action-Take urgent action to combat climate change and its impacts.

SDG 14 – Life Below Water-Conserve and sustainably use the oceans, seas, and marine resources.

SDG 15 – Life on Land -Protect, restore, and promote sustainable use of terrestrial ecosystems, manage forests sustainably, combat desertification, halt and reverse land degradation, and halt biodiversity loss.

SDG 16 – Peace, Justice, and Strong Institutions- Promote peaceful and inclusive societies, provide access to justice for all, and build effective, accountable, and inclusive institutions.

SDG 17 – Partnerships for the Goals -Strengthen the means of implementation and revitalize the global partnership for sustainable development

MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR

DEPARTMENT OF COMPUTER APPLICATIONS

APRIL 2026

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DECLARATION

I hereby certify that the work which is being presented in the project entitled "DIABETIC RETINOPATHY SEVERITY CLASSIFICATION USING VISION TRANSFORMER" submitted in the ***DEPARTMENT OF COMPUTER APPLICATIONS*** *is an authentic record of my own work carried under the supervision of* ***MS. SREELEKSHMI K S, ASSISTANT PROFESSOR.*** *This study has not been submitted to any other institution or university for the award of any other degree. This report has been checked for plagiarism by the college and the similarity index is within permissible limits set by the college.*

Name & Signature of Student

Date:

Place

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CERTIFICATE

This is to certify that the Project titled ***"DIABETIC RETINOPATHY SEVERITY CLASSIFICATION USING VISION TRANSFORMER"*** *is the bonafide record of the work done by* ***ARJUN AJAYAKUMAR (MLM24MCA-2016)*** *of Master of Computer Applications towards the partial fulfilment of the requirement for the award of MASTER OF COMPUTER APPLICATIONS* ***by APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY****, during the academic year 2025-26.*

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ABSTRACT

Diabetic Retinopathy is a serious eye disease caused by diabetes that can lead to vision loss or blindness if not detected at an early stage. Regular screening of retinal images is important for timely diagnosis and treatment. However, manual examination of retinal fundus images by ophthalmologists can be time-consuming and may not always be available in all healthcare facilities. The Automated Diabetic Retinopathy Detection System provides a smart solution to assist in early diagnosis using deep learning techniques.

The system uses retinal fundus images as input and analyzes them using a trained deep learning model to detect the presence and severity of diabetic retinopathy. It helps in identifying different stages of the disease and provides quick and accurate results. The platform allows healthcare professionals to upload retinal images, view prediction results, and generate reports through an easy-to-use interface.

By using modern technologies such as deep learning, image processing, and web-based applications, the system improves the efficiency of medical diagnosis and reduces the workload of specialists. This automated approach supports early detection, faster screening, and better patient care, especially in areas where expert ophthalmologists are limited. The system also contributes to improving healthcare services through the use of intelligent medical technology.

Keywords:

Diabetic Retinopathy Detection, Deep Learning, Retinal Image Analysis, Medical Image Processing, Automated Diagnosis SystemSecure Login System, Classroom Monitoring, Notification System, User-Friendly Interface,

Mapping with Sustainable Development Goals	SDG 3 - Good Health and Well-being SDG 9 -Industry, Innovation and Infrastructure SDG 10 -Reduced Inequalities

CONTENTS

CONTENTS	Title	Page No
	ACKNOWLEDGEMENT	**I**
	ABSTRACT	**II**
	LIST OF FIGURES	**III**
	LIST OF ABBREVIATIONS	**IV**
Chapter 1:	**INTRODUCTION**	**1**
1.1	Background	1
1.2	Motivation	2
1.3	Scope and Objectives	2
1.4	Problem Statement	3
Chapter 2:	**REQUIREMENT AND ANALYSIS**	**6**
2.1	Existing system & Problem Statement	6
2.1	Solution Overview	6
2.3	Feasibility Study	8
2.3.1	Technical feasibility	8
2.3.1	Economic Feasibility	9
2.3.3	Operational Feasibility	9
3	**LITERATURE REVIEW**	**11**
3.1	Vision Transformer with Masked Autoencoders for Referable Diabetic Retinopathy Detection [Yang Yang, Yufei Zhang, Qian Zhao, PLOS ONE Journal, 2024]	11
3.2	Diabetic Retinopathy Prediction Based on Vision Transformer and Capsule Network [M. Oulhadj, S. Benbelkacem, Computers in Biology and Medicine Journal, 2024]	12
3.3	HybridFusionNet: Vision Transformer Based Hybrid Model for Diabetic Retinopathy Detection [A. Shukla, R. Verma, Technologies Journal, 2024]	12
3.4	Deep Learning Based Model for Diabetic Retinopathy Grading [Samia Akhtar, M. Rahman, Scientific Reports Journal, 2025]	13

3.5	Convolutional Neural Network Based Diabetic Retinopathy Classification [R. S. Kavitha, P. Anitha, Applied Sciences Journal, 2024]	15
3.6	Hybrid Deep Learning Framework for Early Detection of Diabetic Retinopathy [Li Wang, Chen Liu, PMC Medical Imaging Journal, 2024]	16
3.7	Vision Transformer Based Automated Detection of Diabetic Retinopathy [Zhou Wei, Li Chen, World Scientific Journal, 2024]	17
3.8	Automated Fundus Image Analysis for Diabetic Retinopathy Detection [Ish Kumar Gupta, Rakesh Sharma, MethodsX Journal, 2025]	17
3.9	Multimodal Deep Learning for Diabetic Retinopathy Detection [Michael Brown, Sarah Johnson, Medical Image Analysis Review, 2025]	18
3.10	Knowledge-Enhanced Transformer Framework for Diabetic Retinopathy Diagnosis [Anirban Samanta, Rajiv Kumar, ArXiv Research Paper, 2025]	18
Chapter 4:	**DIAGRAMS**	**20**
4.1	Data Flow Diagrams	20
4.2	Use Case Diagram	24
4.3	Class Diagram	26
4.4	State Diagram	29
4.5	Activity Diagram	32
4.6	Sequence Diagram	35
Chapter 5:	**METHODOLOGY**	**39**
5.1	Module Descriptions	39
5.2	Algorithms	48
5.3	System Architecture	50
Chapter 6:	**IMPLEMENTATION DETAILS**	**53**
6.1	Implementation Procedures	53
6.2	Testing Methods and Results	60

Chapter 7:	**ADVANTAGES AND DISADVANTAGES**	**64**
7.1	Advantages	64
7.2	Disadvantages	67
Chapter 8:	**RESULT AND DISCUSSION**	**71**
8.1	System Output Overview`	71
8.2	Prediction Results	72
8.3	Feature Detection and Abnormality Identification	73
8.4	Disease Severity Index	74
8.5	Retinal feature visulization	74
Chapter 9:	**CONCLUSION**	**75**
9.1	Summary	75
9.2	Limitations	76
9.3	Future Scope	77
9.4	Application	78
10	**APPENDICES**	**79**
11	**REFERENCES**	**83**

LIST OF FIGURES

FIGURE NO	TITLE	PAGE NO
4.1	DFD Level 0	21
4.2	DFD Level 1	23
4.3	Use case diagram	26
4.4	Class Diagram	28
4.5	State Diagram	30
4.6	Activity Diagram	34
4.7	Sequence Diagram	36
5.1	Home	41
5.2	Image Upload	42
5.3	Result	44
5.4	Report Generation	45
5.5	Admin Dashboad	47
5.6	Patient Diagnostic Module	48
5.7	Architecture	51

LIST OF ABBREVIATIONS

ABBREVIATION		FULL FORM
CNN	-	Convolutional Neural Network
AI	-	Artificial Intelligence
ML	-	Machine Learning
DR	-	Diabetic Retinopathy
DL	-	Deep Learning
GPU	-	Graphics Processing Unit
API	-	Application Programming Interface
CAD	-	Computer Aided Diagnosis

CHAPTER 1
INTRODUCTION

1.1 Background

Diabetes mellitus is one of the most common chronic diseases worldwide. One of the most severe complications of diabetes is Diabetic Retinopathy (DR), which affects the retina of the eye and may eventually lead to permanent blindness if not detected at an early stage.The retina is a light-sensitive layer located at the back of the eye that converts light into neural signals which are interpreted by the brain as visual images. In diabetic patients, high blood sugar levels can damage the small blood vessels of the retina. Over time, these damaged vessels leak fluid or blood, causing swelling, blockage, or abnormal blood vessel growth. This condition is known as Diabetic Retinopathy**.**According to the World Health Organization (WHO)**,** diabetic retinopathy is one of the leading causes of vision loss globally. Millions of people suffer from this disease every year, and the number continues to grow due to the increasing prevalence of diabetes.Traditionally, diabetic retinopathy is diagnosed manually by ophthalmologists using retinal fundus images. However, manual examination has several limitations:

• It requires experienced specialists
• The process is time-consuming
• Diagnosis may vary depending on the doctor’s experience
• Large-scale screening is difficult

With the advancement of Artificial Intelligence (AI) and Deep Learning, automated systems can now analyze retinal images and detect diseases with high accuracy.Recent research has introduced Vision Transformer (ViT) models that can effectively analyze medical images. Vision Transformers use attention mechanisms to capture complex visual patterns in images, making them suitable for tasks such as diabetic retinopathy classification.This project proposes an automated system that uses Vision Transformer models to classify diabetic retinopathy severity levels from retinal images.

1.2 Motivation

The motivation behind developing an automated diabetic retinopathy detection system arises from several healthcare challenges.The number of people suffering from diabetes is increasing rapidly across the world. As a result, the number of patients at risk of developing diabetic retinopathy is also increasing.Manual screening of retinal images requires trained ophthalmologists and specialized equipment. However, many rural and underdeveloped regions face a shortage of medical experts.Another important motivation is the need for early detection. In many cases, diabetic retinopathy does not show noticeable symptoms in its early stages. Patients often realize the problem only after vision damage has already occurred.Automated detection systems can assist doctors by analyzing retinal images quickly and accurately. These systems can screen large numbers of patients and identify those who require immediate medical attention.The use of artificial intelligence in medical image analysis can significantly improve diagnostic efficiency and reduce the workload of healthcare professionals.Therefore, the development of an automated diabetic retinopathy detection system using Vision Transformer technology can greatly improve early diagnosis and healthcare accessibility.

1.3 Scope & Objectives

Diabetic Retinopathy (DR) is one of the most serious complications of diabetes and a leading cause of vision loss worldwide. It occurs when prolonged high blood sugar levels damage the small blood vessels in the retina, the light-sensitive layer at the back of the eye responsible for vision. These damaged blood vessels may swell, leak fluid, or become blocked, eventually leading to abnormal blood vessel growth and severe vision problems. If diabetic retinopathy is not detected and treated at an early stage, it can result in permanent blindness.

Early diagnosis of diabetic retinopathy is extremely important, but in many cases the disease does not show noticeable symptoms during its initial stages. Patients may not experience any vision problems until the disease becomes severe. Because of this, regular retinal screening is necessary for diabetic patients.

Traditionally, diabetic retinopathy is diagnosed through manual examination of retinal fundus images by ophthalmologists. Specialists analyze the retinal images to identify abnormalities

such as microaneurysms, hemorrhages, exudates, and abnormal blood vessel growth. Although this method is effective, it has several limitations. The diagnosis process is time-consuming and requires experienced medical professionals. In many regions, especially rural areas, there is a shortage of trained ophthalmologists and proper medical infrastructure. As a result, many patients do not receive timely screening and diagnosis.

Another challenge is that manual diagnosis can sometimes be subjective, as different doctors may interpret retinal images differently depending on their experience. Additionally, large-scale screening programs become difficult when thousands of retinal images need to be examined manually.

With the rapid advancement of artificial intelligence and deep learning technologies, automated medical image analysis systems have become possible. Deep learning models can analyze retinal images and identify disease patterns with high accuracy. Among these technologies, Vision Transformer (ViT) models have recently shown strong performance in image classification tasks. Vision Transformers use attention mechanisms to analyze visual features in images and capture complex relationships between different regions of an image.

Therefore, there is a need for an automated system that can analyze retinal fundus images and accurately classify the severity of diabetic retinopathy. The proposed system aims to develop a deep learning-based model using Vision Transformer architecture to detect and classify diabetic retinopathy severity levels. Such a system can assist doctors in diagnosis, reduce workload, and enable large-scale screening of diabetic patients, ultimately helping to prevent vision loss.

1.4 Problem Statement

The primary objective of this project is to design and develop an automated system capable of detecting and classifying the severity of Diabetic Retinopathy (DR) using retinal fundus images. With the rapid advancement of artificial intelligence and deep learning technologies, automated medical image analysis systems can assist healthcare professionals in diagnosing diseases more efficiently and accurately. The proposed system uses a Vision Transformer model to analyze retinal images and identify the stage of diabetic retinopathy present in the image. The detailed objectives of the project are explained below.One of the main objectives

of the project is to create an automated system that can detect diabetic retinopathy from retinal images without requiring manual examination by ophthalmologists. Traditional diagnosis methods depend on trained medical professionals who manually analyze retinal images. This process is time-consuming and may not be feasible for large-scale screening programs. By developing an automated detection system, the project aims to provide a faster and more efficient solution for identifying diabetic retinopathy in patients.Early detection plays a crucial role in preventing vision loss caused by diabetic retinopathy. In many cases, the disease does not show visible symptoms during its early stages, making it difficult for patients to recognize the condition. An automated system that analyzes retinal images can detect subtle changes in the retina that may indicate the presence of diabetic retinopathy. Early detection allows doctors to provide timely treatment and prevent the progression of the disease.

Another important objective of the project is to explore the use of Vision Transformer (ViT) architecture for retinal image classification. Vision Transformers are a recent advancement in deep learning that use attention mechanisms to analyze images. Unlike traditional convolutional neural networks, Vision Transformers divide images into patches and process them using transformer-based architectures. This approach enables the model to capture complex patterns and relationships within the image, improving classification accuracy.Retinal images obtained from fundus cameras may contain noise, variations in brightness, and differences in image resolution. These factors can affect the performance of the deep learning model. Therefore, another objective of the project is to apply image preprocessing techniques to improve image quality before feeding them into the model. Preprocessing steps include image resizing, normalization, contrast enhancement, and noise removal. These steps help the model extract meaningful features from the retinal images.The proposed system aims to classify retinal images into different severity levels of diabetic retinopathy. The classification is based on internationally accepted standards used in medical diagnosis. The system will categorize images into five classes: No Diabetic Retinopathy, Mild Diabetic Retinopathy, Moderate Diabetic Retinopathy, Severe Diabetic Retinopathy, and Proliferative Diabetic Retinopathy. This classification helps doctors determine the stage of the disease and decide the appropriate treatment plan for the patient.The project also aims to develop a simple and user-friendly interface that allows users to upload retinal images and obtain classification results.

The interface will be designed in such a way that healthcare professionals can easily interact with the system without requiring technical knowledge of machine learning. The interface will display the predicted class and the confidence score of the prediction.Another important objective is to evaluate the performance of the Vision Transformer model using standard evaluation metrics. These metrics include accuracy, precision, recall, and F1-score. Performance evaluation helps determine how effectively the model can classify retinal images and detect diabetic retinopathy.

Finally, the project aims to support large-scale screening of diabetic patients. Automated systems can analyze thousands of retinal images in a short period of time, making them suitable for mass screening programs. This can help healthcare organizations identify patients who require further medical examination and treatment.

CHAPTER 2

REQUIREMENT AND ANALYSIS

2.1 Existing System & Problem Statement

With the rapid increase in the number of diabetic patients worldwide, diabetic retinopathy has become one of the leading causes of vision loss and blindness. Diabetic retinopathy occurs when prolonged high blood sugar levels damage the small blood vessels in the retina, which is the light-sensitive tissue located at the back of the eye.

Currently, the detection of diabetic retinopathy is mainly performed through manual examination by ophthalmologists. Doctors analyze retinal fundus images obtained from fundus cameras to identify abnormalities such as microaneurysms, hemorrhages, exudates, and abnormal blood vessel growth.

Although this traditional method is widely used in hospitals and clinics, it has several limitations.

One of the major limitations is the dependence on trained ophthalmologists. Accurate diagnosis requires experienced specialists who can carefully examine retinal images. However, in many rural or underdeveloped regions, access to ophthalmologists is limited.

Another major limitation is the time-consuming nature of manual diagnosis. Each retinal image must be examined carefully by a medical expert, which makes it difficult to screen a large number of patients in a short time. As the number of diabetic patients increases, healthcare systems face challenges in providing timely screening.

Manual diagnosis can also be subjective, as different doctors may interpret the same retinal image differently depending on their experience and expertise. This may lead to inconsistencies in diagnosis.

Furthermore, early stages of diabetic retinopathy may not produce noticeable symptoms. Patients often remain unaware of the disease until it progresses to an advanced stage. Because of this, many cases are detected late, when treatment becomes more difficult.

These limitations highlight the need for an automated system that can analyze retinal images efficiently and assist doctors in diagnosing diabetic retinopathy at an early stage**.**

2.2 Solution Overview

To overcome the limitations of the existing system, the proposed project introduces an automated diabetic retinopathy detection and severity classification system using Vision Transformer models.

The proposed system is designed to automatically analyze retinal fundus images and classify them into different severity levels of diabetic retinopathy. The system uses deep learning techniques to identify patterns and abnormalities in retinal images that indicate the presence of the disease.The system works through several stages.

First, retinal images are provided as input by the user through a web interface. These images are captured using retinal fundus cameras and uploaded to the system.

After receiving the input image, the system performs image preprocessing. During this stage, the image is resized, normalized, and enhanced to improve its quality. Preprocessing helps remove noise and improve the visibility of important retinal features.

Once preprocessing is completed, the image is passed to the Vision Transformer model. Vision Transformers divide images into smaller patches and analyze them using attention mechanisms. This allows the model to capture complex visual patterns and relationships within the retinal image.

The deep learning model then extracts important features from the image and analyzes them to detect abnormalities associated with diabetic retinopathy.

Based on the extracted features, the system classifies the retinal image into one of the following five severity levels:

•No
•Mild
•Moderate
•Severe
• Proliferative

After classification, the system displays the predicted severity level along with a confidence score indicating the reliability of the prediction.

The results are presented through a simple and user-friendly interface that allows healthcare professionals to easily interpret the output.

By combining deep learning technology with retinal image analysis, the proposed system provides a faster and more efficient solution for detecting diabetic retinopathy.

2.3 Feasibility Study

A feasibility study is conducted to determine whether the proposed system can be successfully developed and implemented. The feasibility analysis evaluates different aspects such as technical feasibility, economic feasibility, and operational feasibility.

2.3.1 Technical Feasibility

Technical feasibility evaluates whether the required technology and resources are available to implement the proposed system.

The proposed diabetic retinopathy detection system can be developed using widely available technologies and machine learning frameworks.

Programming languages such as Python provide strong support for deep learning and image processing tasks. Several libraries and frameworks are available that simplify the development process.

For example:

•TensorFlow and PyTorch can be used to implement deep learning models.
•OpenCV
•NumPy
• Flask or Streamlit can be used to develop the web interface.

The Vision Transformer model can be trained using publicly available retinal image datasets such as the Kaggle EyePACS dataset.

Because these tools and technologies are widely available and well documented, the development of the proposed system is technically feasible.

2.3.2 Economic Feasibility

Economic feasibility evaluates the cost involved in developing and maintaining the proposed system.

The proposed system can be developed using open-source software and libraries, which significantly reduces development costs. Most of the required tools such as Python, TensorFlow, and OpenCV are freely available.

The system does not require expensive hardware infrastructure. A standard computer with a GPU can be used to train the deep learning model.

In addition, once the system is developed, it can analyze large numbers of retinal images automatically without requiring continuous manual supervision. This can reduce the cost of screening diabetic patients.Therefore, the project is economically feasible and suitable for research and healthcare applications.

2.3.3 Operational Feasibility

Operational feasibility evaluates how easily the system can be used by its intended users.

The proposed system is designed with a simple and user-friendly interface that allows healthcare professionals to upload retinal images and obtain diagnostic results quickly.

Users do not need advanced technical knowledge to operate the system. The interface provides clear instructions for uploading images and viewing prediction results.

The system automatically performs all processing steps including preprocessing, feature extraction, and classification.

The results are displayed in an easy-to-understand format, showing the predicted severity level and confidence score.

Because the system is simple to use and requires minimal training, it is operationally feasible for use in hospitals, clinics, and research environments.

CHAPTER 3
LITERATURE REVIEW

3.1 Vision Transformer with Masked Autoencoders for Referable Diabetic Retinopathy Detection [Yang Yang, Yufei Zhang, Qian Zhao, PLOS ONE Journal, 2024]

Diabetic retinopathy is one of the leading causes of blindness worldwide, and early detection is crucial for preventing severe vision loss. Traditional diagnosis relies on manual examination of retinal images by ophthalmologists. However, this process is time-consuming and requires highly skilled medical professionals. With the rapid advancement of artificial intelligence, automated deep learning-based systems have become an effective solution for detecting diabetic retinopathy from retinal images.

Yang et al. proposed a deep learning framework based on the Vision Transformer architecture for detecting referable diabetic retinopathy. Vision Transformer models divide input images into smaller patches and process them using transformer layers that employ attention mechanisms. Unlike traditional convolutional neural networks, Vision Transformers are capable of capturing global relationships between different regions of an image.

In this research, the authors introduced masked autoencoder techniques to improve the feature extraction capability of the Vision Transformer model. The masked autoencoder reconstructs missing image patches during training, allowing the model to learn meaningful representations of retinal images. The system was trained on a large dataset of retinal fundus images and evaluated for classification performance.

Experimental results demonstrated that the Vision Transformer model achieved competitive performance in detecting referable diabetic retinopathy. The model successfully identified abnormal retinal patterns associated with the disease and provided reliable classification results. The study concluded that transformer-based models have strong potential for developing automated medical diagnostic systems for diabetic retinopathy screening.

3.2 Diabetic Retinopathy Prediction Based on Vision Transformer and Capsule Network [M. Oulhadj, S. Benbelkacem, Computers in Biology and Medicine Journal, 2024]

The increasing prevalence of diabetes has led to a growing need for efficient diagnostic tools for detecting diabetic retinopathy. Manual analysis of retinal images is not only time-consuming but also prone to human error. To address this challenge, researchers have explored advanced deep learning techniques capable of automatically analyzing retinal images.

Oulhadj et al. proposed a hybrid deep learning architecture that combines Vision Transformer models with capsule networks for diabetic retinopathy detection. The Vision Transformer component is responsible for extracting global features from retinal images by analyzing relationships between image patches. Capsule networks, on the other hand, capture hierarchical relationships between features and improve classification accuracy.

The hybrid model processes retinal images through several stages including image preprocessing, feature extraction, and classification. During preprocessing, retinal images are resized, normalized, and enhanced to improve the quality of the input data. The Vision Transformer then extracts high-level features, which are passed to the capsule network for classification.

Experimental results showed that the hybrid architecture significantly improved classification accuracy compared to traditional convolutional neural networks. The system demonstrated strong performance in detecting different severity levels of diabetic retinopathy. The study highlights the effectiveness of combining transformer-based architectures with capsule networks for medical image analysis.

3.3 HybridFusionNet: Vision Transformer Based Hybrid Model for Diabetic Retinopathy Detection [A. Shukla, R. Verma, Technologies Journal, 2024]

Deep learning models have shown remarkable success in medical image classification tasks. However, many models struggle to capture both local and global features simultaneously. Convolutional neural networks are effective at extracting local features, while transformer architectures are better at capturing global relationships within images.

Shukla et al. proposed a hybrid deep learning model called HybridFusionNet to address this limitation. The model integrates convolutional neural networks with Vision Transformer architecture to analyze retinal fundus images. CNN layers are used to extract local visual

features such as microaneurysms and hemorrhages, while transformer layers capture global contextual information.

The proposed system processes retinal images through multiple layers of convolutional and transformer blocks. Feature maps generated by CNN layers are combined with transformer embeddings to improve classification performance. This hybrid approach enables the model to learn both detailed and contextual features of retinal images.

Experimental evaluation of the model demonstrated high classification accuracy in detecting diabetic retinopathy severity levels. The hybrid architecture significantly outperformed traditional CNN models and achieved better generalization across different datasets. The study concluded that combining CNN and Vision Transformer architectures can improve the reliability of automated diabetic retinopathy detection systems.

3.4 Deep Learning Based Model for Diabetic Retinopathy Grading [Samia Akhtar, M. Rahman, Scientific Reports Journal, 2025]

Diabetic retinopathy is one of the most common complications of diabetes and a major cause of vision impairment worldwide. Early detection and accurate classification of diabetic retinopathy are essential for preventing permanent vision loss. Traditional diagnosis methods rely on manual examination of retinal fundus images by ophthalmologists. Although manual diagnosis is effective, it requires significant expertise and time. As the number of diabetic patients continues to increase globally, there is a growing need for automated diagnostic systems that can analyze retinal images quickly and accurately.

Akhtar et al. proposed a deep learning-based model for grading diabetic retinopathy severity using retinal fundus images. The primary objective of this research was to develop an automated system capable of detecting diabetic retinopathy and classifying it into different severity levels. The authors emphasized that early identification of diabetic retinopathy can significantly reduce the risk of blindness through timely medical intervention.

In the proposed approach, retinal fundus images were first collected from publicly available datasets. These datasets contained thousands of labeled retinal images categorized according to different stages of diabetic retinopathy. Before training the deep learning model, the images were preprocessed to improve their quality and remove unwanted noise. The preprocessing stage included operations such as image resizing, normalization, contrast enhancement, and

noise reduction. These steps helped improve the clarity of retinal structures and allowed the model to extract more meaningful features.

After preprocessing, the retinal images were fed into a deep learning architecture based on convolutional neural networks. CNN models are particularly well suited for image analysis tasks because they automatically learn spatial features from images. The convolutional layers extract visual patterns such as blood vessel abnormalities, microaneurysms, hemorrhages, and exudates, which are common indicators of diabetic retinopathy.

The proposed model consisted of multiple convolutional layers followed by pooling layers and fully connected layers. The convolutional layers were responsible for extracting features from retinal images, while pooling layers reduced the dimensionality of the feature maps. Fully connected layers were used for classification of the extracted features into different diabetic retinopathy severity levels.

To improve the robustness of the model, the researchers applied data augmentation techniques during training. Data augmentation involved generating additional training samples by applying transformations such as image rotation, horizontal flipping, vertical flipping, scaling, and brightness adjustment. These techniques helped increase the diversity of the training dataset and prevented the model from overfitting.

The model was trained using a supervised learning approach where each retinal image was labeled with its corresponding diabetic retinopathy severity level. During training, the model learned to associate specific visual patterns with different disease stages. The training process involved minimizing a loss function that measured the difference between predicted labels and actual labels.

To evaluate the performance of the proposed system, several evaluation metrics were used. These included accuracy, precision, recall, and F1-score. Accuracy measures the overall correctness of the model's predictions, while precision indicates the proportion of correctly identified positive cases. Recall measures the ability of the model to identify all relevant cases of diabetic retinopathy, and the F1-score provides a balanced measure of precision and recall.

Experimental results demonstrated that the deep learning model achieved high classification accuracy for diabetic retinopathy grading. The model successfully detected retinal abnormalities and classified images into different severity levels with high reliability. The study showed that automated deep learning systems can significantly assist ophthalmologists in diagnosing diabetic retinopathy.

Another important contribution of this research is the potential for large-scale screening of diabetic patients. Automated systems can analyze thousands of retinal images in a short period of time, which is particularly useful for mass screening programs. This can help healthcare providers identify patients who require further medical examination and treatment.

The authors concluded that deep learning-based diabetic retinopathy detection systems have strong potential for improving early diagnosis and reducing the burden on healthcare professionals. However, they also highlighted several challenges, including the need for large annotated datasets and the importance of ensuring model generalization across different patient populations.

Overall, this research demonstrates that deep learning models can provide an efficient and reliable solution for diabetic retinopathy detection and severity classification. The findings of this study contribute to the development of intelligent medical diagnostic systems capable of assisting doctors in early detection of diabetic retinopathy and preventing vision loss.

3.5 Convolutional Neural Network Based Diabetic Retinopathy Classification [R. S. Kavitha, P. Anitha, Applied Sciences Journal, 2024]

Convolutional Neural Networks (CNN) have been widely used in medical image analysis due to their ability to automatically extract spatial features from images. Kavitha and Anitha proposed a CNN-based system for detecting and classifying diabetic retinopathy severity levels using retinal fundus images. The main objective of this research was to develop an automated system that can assist ophthalmologists in identifying diabetic retinopathy at an early stage.

The authors used a large dataset of retinal images obtained from publicly available databases such as EyePACS and Messidor. These datasets contain thousands of retinal fundus images labeled according to different stages of diabetic retinopathy. Before training the CNN model, the retinal images underwent several preprocessing steps. These steps included image resizing, contrast enhancement, normalization, and noise removal. Image preprocessing improves the visibility of important retinal structures and helps the model extract meaningful features.

The CNN architecture used in this research consisted of multiple convolutional layers followed by pooling layers and fully connected layers. The convolutional layers extract features such as blood vessel patterns, microaneurysms, hemorrhages, and exudates that are commonly associated with diabetic retinopathy. Pooling layers reduce the dimensionality of feature maps and improve computational efficiency.

The extracted features were then passed to fully connected layers for classification. The model classified retinal images into multiple severity levels such as No DR, Mild DR, Moderate DR, Severe DR, and Proliferative DR. Experimental results demonstrated that the CNN model achieved high classification accuracy and effectively detected retinal abnormalities.

The study concluded that CNN-based automated systems can significantly assist medical professionals in diagnosing diabetic retinopathy and can be used for large-scale screening programs.

3.6 Hybrid Deep Learning Framework for Early Detection of Diabetic Retinopathy [Li Wang, Chen Liu, PMC Medical Imaging Journal, 2024]

Wang and Liu proposed a hybrid deep learning framework designed to improve the early detection of diabetic retinopathy. The authors highlighted that early stages of diabetic retinopathy are difficult to detect because the symptoms are often subtle and may not be easily visible to the human eye. Automated systems using advanced deep learning techniques can help identify these subtle patterns in retinal images.

The proposed framework combines Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN). CNN models are effective in extracting spatial features from images, while RNN models are capable of analyzing sequential relationships between features. By combining these two architectures, the system can learn both spatial and contextual information from retinal images.

In the first stage of the framework, retinal fundus images are preprocessed using image enhancement techniques such as histogram equalization and noise filtering. These preprocessing techniques improve the contrast and clarity of the retinal images.

After preprocessing, the CNN layers extract important visual features from the images. These features include retinal lesions, blood vessel abnormalities, and structural changes in the retina. The extracted features are then processed by RNN layers that analyze relationships between different features.

Experimental results showed that the hybrid deep learning model achieved better classification performance compared to traditional CNN-based models. The system was able to detect early signs of diabetic retinopathy with high accuracy.

The authors concluded that hybrid deep learning frameworks provide a powerful approach for automated diabetic retinopathy detection and can significantly improve early diagnosis.

3.7 Vision Transformer Based Automated Detection of Diabetic Retinopathy [Zhou Wei, Li Chen, World Scientific Journal, 2024]

Zhou Wei and Li Chen proposed an automated diabetic retinopathy detection system based on Vision Transformer architecture. Vision Transformers represent a new approach to image classification that utilizes attention mechanisms to analyze relationships between different regions of an image.

Unlike traditional convolutional neural networks, Vision Transformers divide images into small patches and convert these patches into embedding vectors. These embeddings are then processed by transformer layers that analyze relationships between patches using self-attention mechanisms.

In this research, retinal fundus images were divided into smaller patches, and positional encoding was applied to preserve spatial information. The Vision Transformer model then analyzed these patches to extract global contextual features from the entire image.

The model was trained using a large dataset of retinal images labeled according to diabetic retinopathy severity levels. During training, the system learned to recognize important visual patterns such as microaneurysms, hemorrhages, and exudates.

Experimental results demonstrated that the Vision Transformer model achieved competitive classification accuracy compared to traditional CNN-based models. The transformer architecture showed strong capability in capturing global relationships across retinal images.

The study concluded that Vision Transformer models have great potential for developing advanced medical diagnostic systems for diabetic retinopathy detection.

3.8 Automated Fundus Image Analysis for Diabetic Retinopathy Detection [Ish Kumar Gupta, Rakesh Sharma, MethodsX Journal, 2025]

Gupta and Sharma proposed an automated fundus image analysis system for detecting diabetic retinopathy using deep learning techniques. Fundus photography is one of the most widely used methods for capturing retinal images. However, manual analysis of fundus images requires experienced ophthalmologists and can be time-consuming.

The proposed system begins with image preprocessing to improve the quality of retinal images. Preprocessing steps include histogram equalization, contrast enhancement, and removal of background noise. These techniques improve the visibility of retinal structures such as blood vessels and lesions.

After preprocessing, a deep learning model analyzes the retinal images to extract important features. The model identifies abnormalities such as microaneurysms, hemorrhages, and exudates that indicate the presence of diabetic retinopathy.

The extracted features are then used for classification of retinal images into different severity levels. Experimental results showed that the automated system achieved high classification accuracy and was able to detect diabetic retinopathy effectively.

The authors concluded that automated fundus image analysis systems can significantly improve the efficiency of diabetic retinopathy screening and reduce the workload of healthcare professionals.

3.9 Multimodal Deep Learning for Diabetic Retinopathy Detection [Michael Brown, Sarah Johnson, Medical Image Analysis Review, 2025]

Brown and Johnson conducted a comprehensive study on multimodal deep learning approaches for diabetic retinopathy detection. Traditional diagnostic systems rely solely on retinal images for disease detection. However, combining multiple data sources can significantly improve diagnostic accuracy.

The authors proposed a multimodal deep learning framework that integrates retinal images with clinical data such as patient history, blood glucose levels, and laboratory reports. By combining these different sources of information, the system can provide a more comprehensive understanding of the disease.

In the proposed framework, retinal images are analyzed using convolutional neural networks, while clinical data is processed using fully connected neural networks. The outputs from both networks are combined to produce the final prediction.

Experimental results demonstrated that multimodal models achieved higher classification accuracy compared to single-modal systems. The study highlighted the importance of integrating clinical knowledge with image analysis for improving automated diagnosis.

3.10 Knowledge-Enhanced Transformer Framework for Diabetic Retinopathy Diagnosis [Anirban Samanta, Rajiv Kumar, ArXiv Research Paper, 2025]

Samanta and Kumar proposed a knowledge-enhanced transformer framework for diabetic retinopathy diagnosis. The authors recognized that traditional deep learning models rely only on image data and do not utilize additional medical knowledge.

The proposed system integrates retinal fundus images with clinical information using transformer-based architecture. The system uses separate encoders to process different types of input data such as retinal images, medical reports, and patient health records.

The retinal images are processed using a Vision Transformer encoder that extracts visual features from image patches. Clinical information is processed using text encoders that analyze medical reports and patient data.

The extracted features are then combined using a joint transformer network that performs diabetic retinopathy severity classification. This integrated approach allows the system to analyze relationships between visual patterns and clinical information.

Experimental results showed that the knowledge-enhanced transformer model achieved improved classification accuracy compared to traditional deep learning models. The study demonstrated that integrating medical knowledge with deep learning techniques can significantly improve automated disease diagnosis.

CHAPTER 4

DIAGRAMS

System diagrams play an important role in understanding the structure, workflow, and interactions between different components of a software system. These diagrams provide a visual representation of how the Diabetic Retinopathy Detection System operates internally.

The diagrams included in this chapter help developers and users understand how retinal images are processed and analyzed by the system. They illustrate how user input is received, how images are processed using deep learning algorithms, and how classification results are generated.

The system uses several types of diagrams including:

•DataFlowDiagram(DFD)
•UseCaseDiagram
•ClassDiagram
•StateDiagram
•ActivityDiagram
• Sequence Diagram

Each diagram provides a different perspective of the system design and helps explain how the system functions.

4.1 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is used to represent the movement of data within a system. It illustrates how data enters the system, how it is processed, and how the results are generated.

In the Diabetic Retinopathy Detection System, the DFD explains how retinal images are uploaded by the user, processed by the system, analyzed using a Vision Transformer model, and finally classified into different diabetic retinopathy severity levels.

The DFD also identifies important components of the system including:

•Externalentities
•Systemprocesses
•Datastorage
•Data flow between components

DFDs help simplify the understanding of system functionality and provide a clear overview of the system architecture.

4.1.1 DFD Level 0

DFD Level 0 represents the entire system as a single process and shows the interaction between external entities and the system.

In this project, the main external entity is the User, who interacts with the Diabetic Retinopathy Detection System.

The user uploads a retinal image into the system. The system processes the image and returns the prediction result indicating the severity level of diabetic retinopathy.

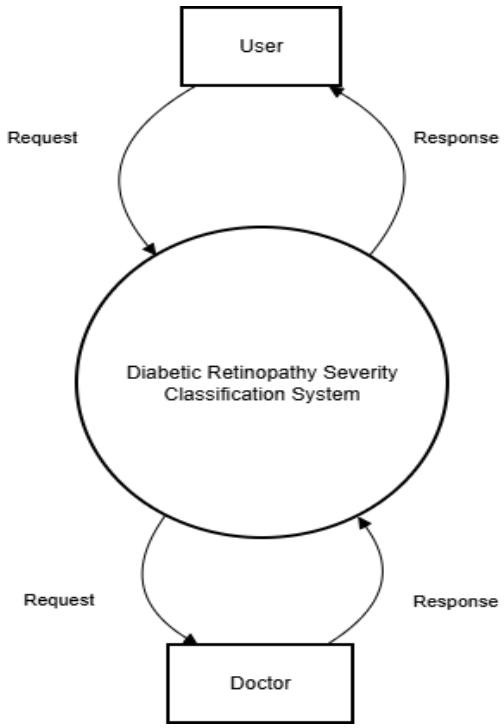


Figure 4.1 DFD Level 0

Components of DFD Level 0

External Entity

• User – The user uploads retinal images for analysis.

System Process

• Diabetic Retinopathy Detection System – This process includes image preprocessing, feature extraction, and classification using the Vision Transformer model.

Input Data

• Retinal fundus image uploaded by the user.

Output Data

•PredictedDRseveritylevel
•Confidencescoreofprediction
•Visualization of detected features

4.1.2 DFD Level 1

DFD Level 1 expands the Level 0 diagram by breaking down the system into multiple processes.

The Diabetic Retinopathy Detection System consists of the following processes:

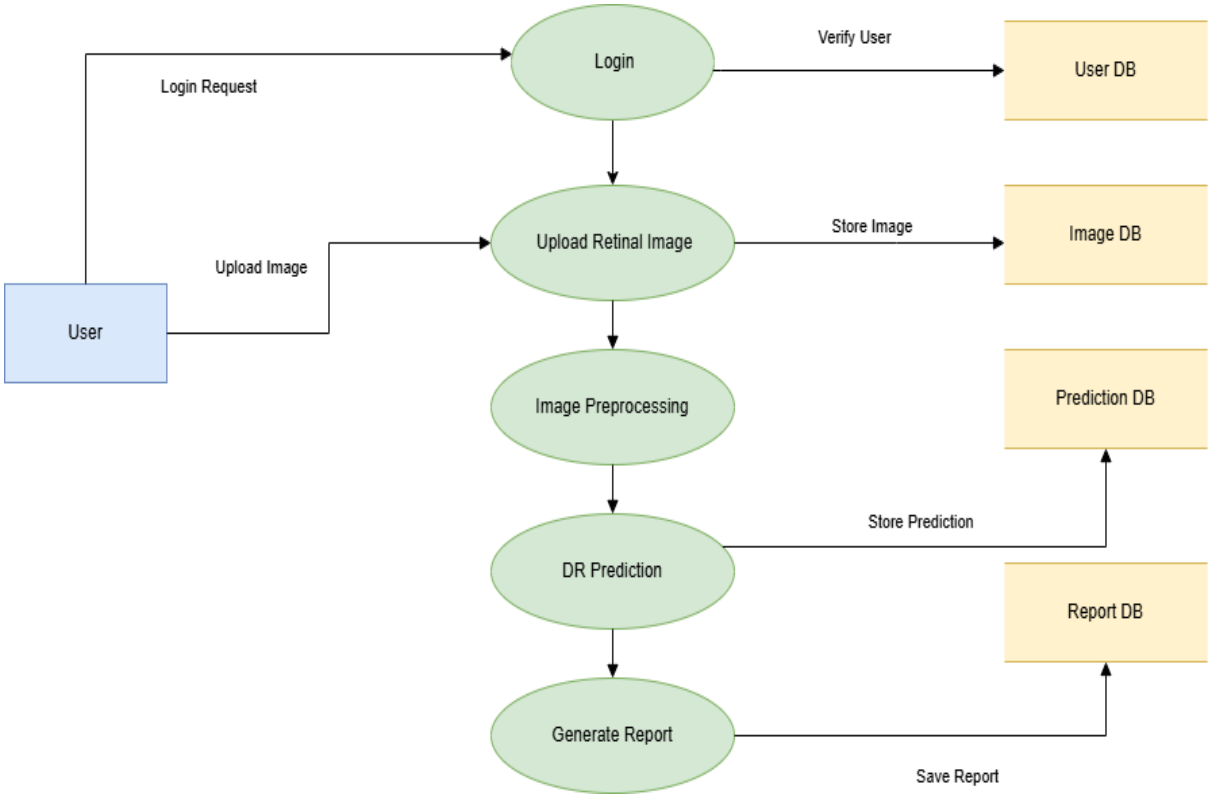


Figure 4.2 DFD Level 1

Process 1: Upload Retinal Image

The user uploads a retinal fundus image through the system interface. The system verifies the uploaded image format and prepares it for processing.

Process 2: Image Preprocessing

Before analysis, the retinal image undergoes preprocessing steps including:

• Image resizing

• Noise removal

• Contrast enhancement

• Normalization

These steps improve image quality and help the model extract meaningful features.

Process 3: Feature Extraction

The processed image is divided into smaller patches and converted into embeddings. These embeddings represent important visual features extracted from the retinal image.

Process 4: Vision Transformer Model Analysis

The Vision Transformer model analyzes relationships between image patches using attention mechanisms. The model learns patterns associated with diabetic retinopathy such as microaneurysms, hemorrhages, and exudates.

Process 5: Classification and Result Generation

The system classifies the retinal image into one of the following severity levels:

•No
•Mild
•Moderate
•Severe
• Proliferative

4.2 Use Case Diagram

A Use Case Diagram represents the interaction between the user and the system. It describes the various operations that a user can perform within the Diabetic Retinopathy Detection System. The purpose of the use case diagram is to identify the functionalities of the system and show how users interact with different system components.

In the proposed system, the primary actor is the User, who interacts with the system through a web interface. The user uploads retinal fundus images obtained from a retinal camera or medical imaging device. After uploading the image, the system processes the image using preprocessing techniques and deep learning algorithms to detect the severity level of diabetic retinopathy.

The use case diagram includes several functionalities that represent the main operations of the system. These functionalities include uploading retinal images, preprocessing the images, analyzing the images using the Vision Transformer model, predicting the severity level of diabetic retinopathy, and displaying the results to the user.

The system first allows the user to upload a retinal image through the interface. Once the image is uploaded, the system verifies whether the image format is valid and suitable for analysis. After validation, the image is passed to the preprocessing module.

The image preprocessing stage prepares the retinal image for analysis by applying operations such as resizing, normalization, noise removal, and contrast enhancement. These preprocessing techniques improve the quality of the retinal image and highlight important features such as blood vessels and lesions.

After preprocessing, the image is passed to the **Vision** Transformer model, which analyzes the image to detect abnormalities associated with diabetic retinopathy. The model extracts features from the retinal image and compares them with patterns learned during the training process.

The system then predicts the severity level of diabetic retinopathy based on the features extracted from the retinal image. The prediction is classified into one of the following categories: No Diabetic Retinopathy, Mild Diabetic Retinopathy, Moderate Diabetic Retinopathy, Severe Diabetic Retinopathy, or Proliferative Diabetic Retinopathy.

Finally, the prediction result is displayed to the user along with the confidence score of the model. This result helps medical professionals understand the severity level of the disease and determine appropriate treatment options.

The use case diagram provides a clear representation of how the user interacts with the system and how different system functions work together to detect diabetic retinopathy.

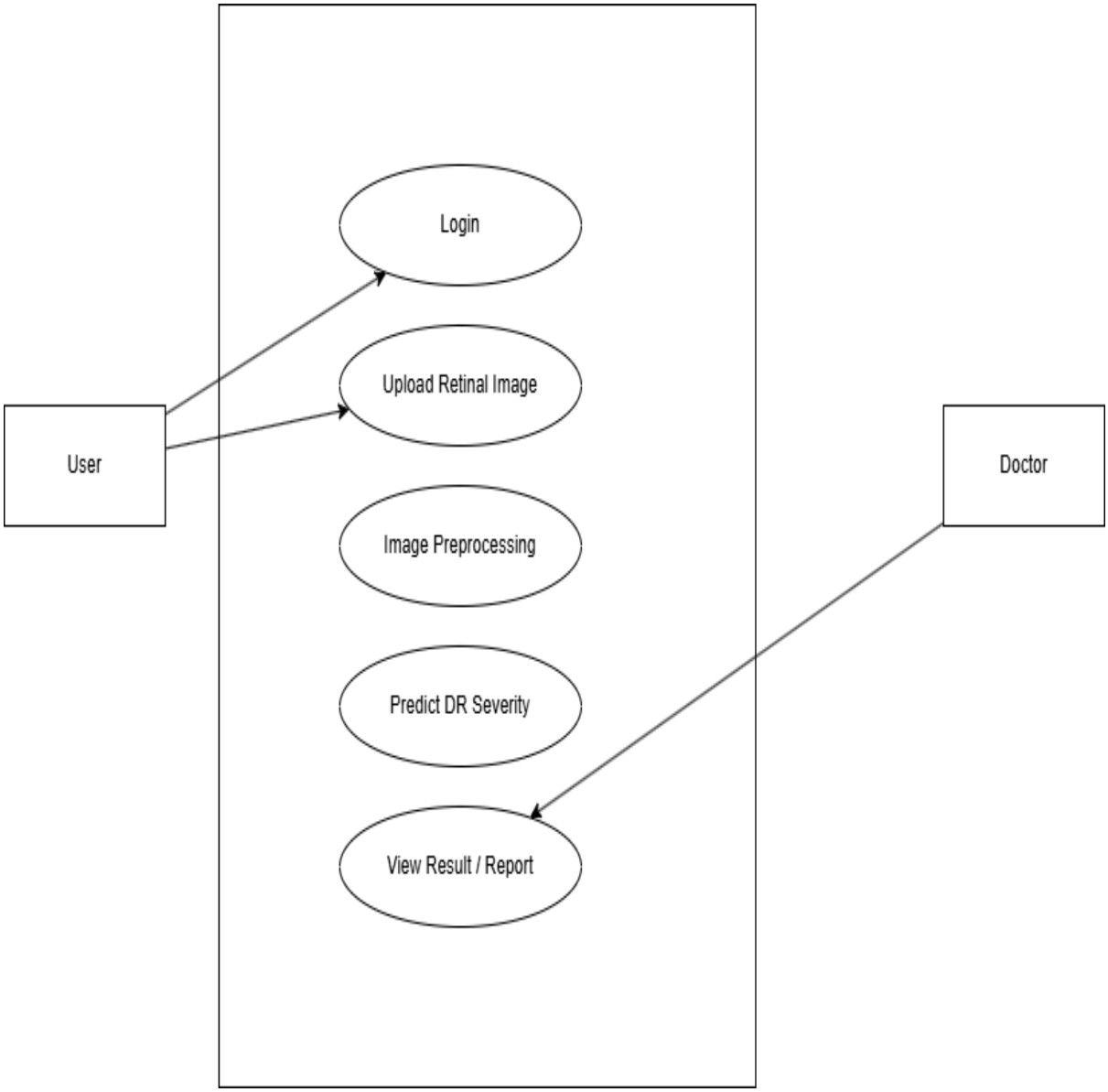


Figure 4.3 Use case diagram

4.3 Class Diagram

This use case allows the user to upload a retinal fundus image into the system interface. The uploaded image is the primary input used for detecting diabetic retinopathy.

• **Validate Image Input**

The image is uploaded, the system verifies whether the file format and image quality suitable for analysis. If the image is invalid or corrupted, the system displays an error message.

• **Preprocess Retinal Image**

The system performs image preprocessing operations to improve the quality of the retinal image. These operations include image resizing, noise removal, contrast enhancement, and normalization to prepare the image for analysis.

• **Analyze Retinal Image**

This is the main functionality of the system. The Vision Transformer model analyzes the retinal image and extracts important visual features related to diabetic retinopathy such as microaneurysms, hemorrhages, and exudates.

• **Predict DR Severity Level**

After analyzing the retinal image, the system classifies the image into one of the diabetic retinopathy severity levels. The possible classes include No Diabetic Retinopathy, Mild Diabetic Retinopathy, Moderate Diabetic Retinopathy, Severe Diabetic Retinopathy, and Proliferative Diabetic Retinopathy.

• **View Prediction Result**

After classification, the system displays the predicted severity level of diabetic retinopathy to the user through the interface.

• **View Confidence Score**

The system also provides a confidence score that indicates the probability of the prediction. This helps the user understand how confident the model is about the classification result

• **View Visual Analysis**

The platform may generate visual representations of the analysis results such as highlighted retinal features or detection regions that indicate abnormalities in the retinal image.

• **View Analysis History**

The system stores previous image analyses and prediction results in a history database. Users can review previously analyzed retinal images and their corresponding classification results.

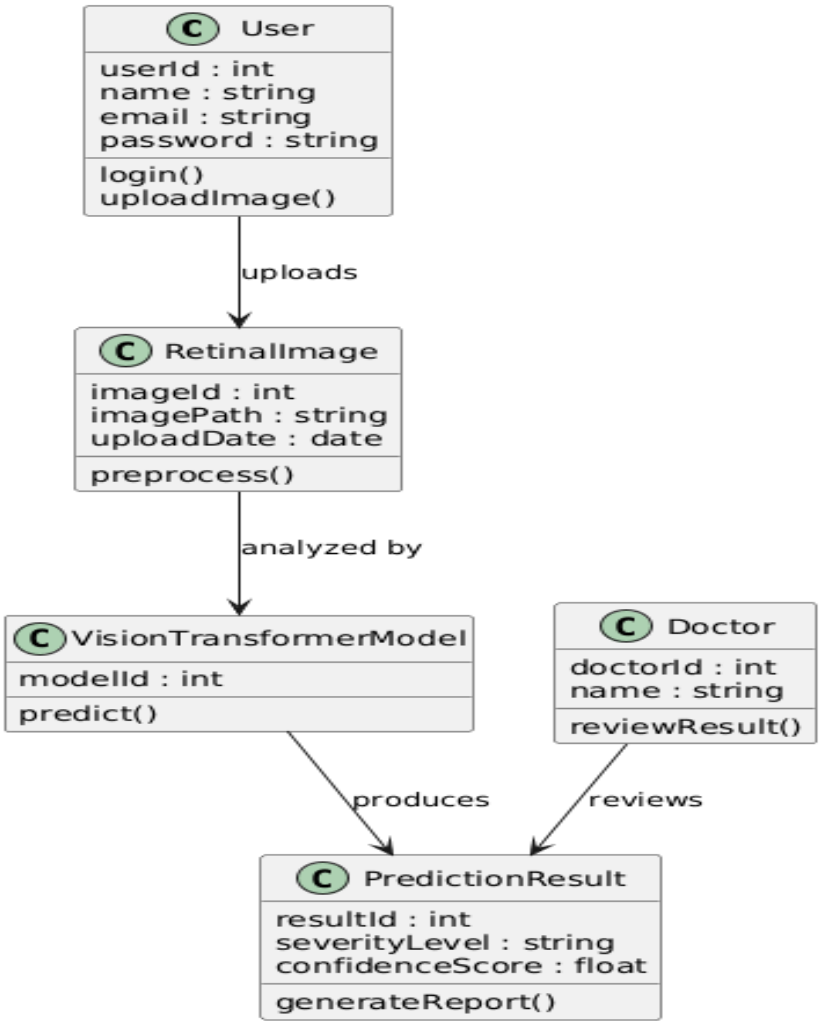


Figure 4.4 Class Diagram

4.4 State Diagram

A State Diagram represents the different states through which the system passes during its execution. It illustrates how the system transitions from one state to another based on user actions or internal processing steps. In the Diabetic Retinopathy Detection System, the state diagram shows the sequence of operations performed when a user uploads a retinal image and the system analyzes it to determine the severity level of diabetic retinopathy.

The state diagram helps in understanding how the system behaves at different stages of the detection process. It also shows the workflow from the initial system start state until the final result is displayed to the user.

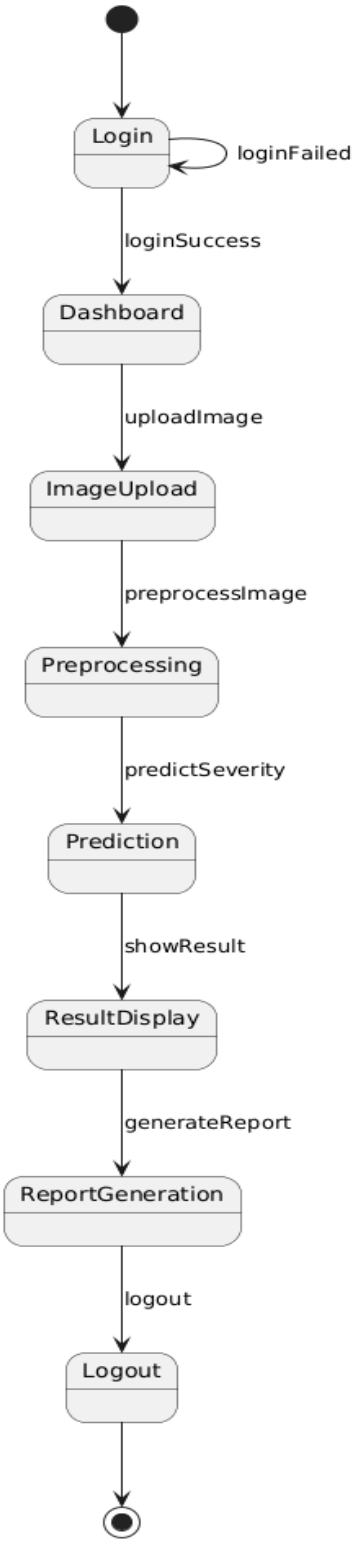


Figure 4.5 State Diagram

• Initial State

The process begins when the system is launched and the application interface is loaded.

• System Interface Loaded

At this stage, the system interface becomes available for user interaction and the system waits for the user to upload a retinal image.

• Upload Retinal Image

The user uploads a retinal fundus image through the system interface. The system receives the image and prepares it for validation.

• Validate Image Input

The system checks whether the uploaded image is valid and in the correct format. If the image is invalid, the system displays an error message and returns to the upload state.

• Preprocess Image

The retinal image undergoes preprocessing operations such as image resizing, noise removal, contrast enhancement, and normalization to improve image quality.

• Feature Extraction

The processed image is divided into patches and converted into embeddings so that the Vision Transformer model can extract meaningful features.

• Model Prediction

The Vision Transformer model analyzes the extracted features and predicts the severity level of diabetic retinopathy**.**

• Result Generated

The system generates the prediction result along with the confidence score.

• Display Result

The final result is displayed to the user through the system interface.

• End State

The process ends after the result is displayed and the user exits the system.

4.5 Activity Diagram

An Activity Diagram illustrates the workflow of the system by showing the sequence of activities performed during the diabetic retinopathy detection process. It provides a clear representation of how different operations are carried out within the system from the moment the user interacts with the application until the final prediction result is displayed. Activity diagrams are useful for understanding the logical flow of the system and identifying how different tasks are executed step by step.

In the Diabetic Retinopathy Detection System, the activity diagram describes how retinal images are uploaded, processed, analyzed using the Vision Transformer model, and classified into different severity levels of diabetic retinopathy. The diagram also shows the decision points where the system validates user input and determines the next step in the workflow.

• Start

The process begins when the user opens the diabetic retinopathy detection application.

• Open System Interface

The system loads the application interface and waits for the user to upload a retinal fundus image.

• Upload Retinal Image

The user uploads a retinal image using the provided interface. The system receives the image as input.

• **Validate Image Input**

The system checks whether the uploaded image is valid and in the correct format. If the image is invalid, an error message is displayed and the user is asked to upload a valid image.

• **Image Preprocessing**

If the image is valid, preprocessing operations are applied to enhance image quality. These operations include image resizing, noise removal, contrast enhancement, and normalization.

• **Feature Extraction**

The preprocessed image is passed to the Vision Transformer model, which extracts important visual features from the retinal image.

• **Model Prediction**

The Vision Transformer model analyzes the extracted features and predicts the severity level of diabetic retinopathy.

• **Generate Result**

The system generates the prediction result along with the confidence score.

• **Display Result**

The predicted diabetic retinopathy severity level and confidence score are displayed to the user through the system interface.

• **End**

The process ends after the user views the result or exits the system.

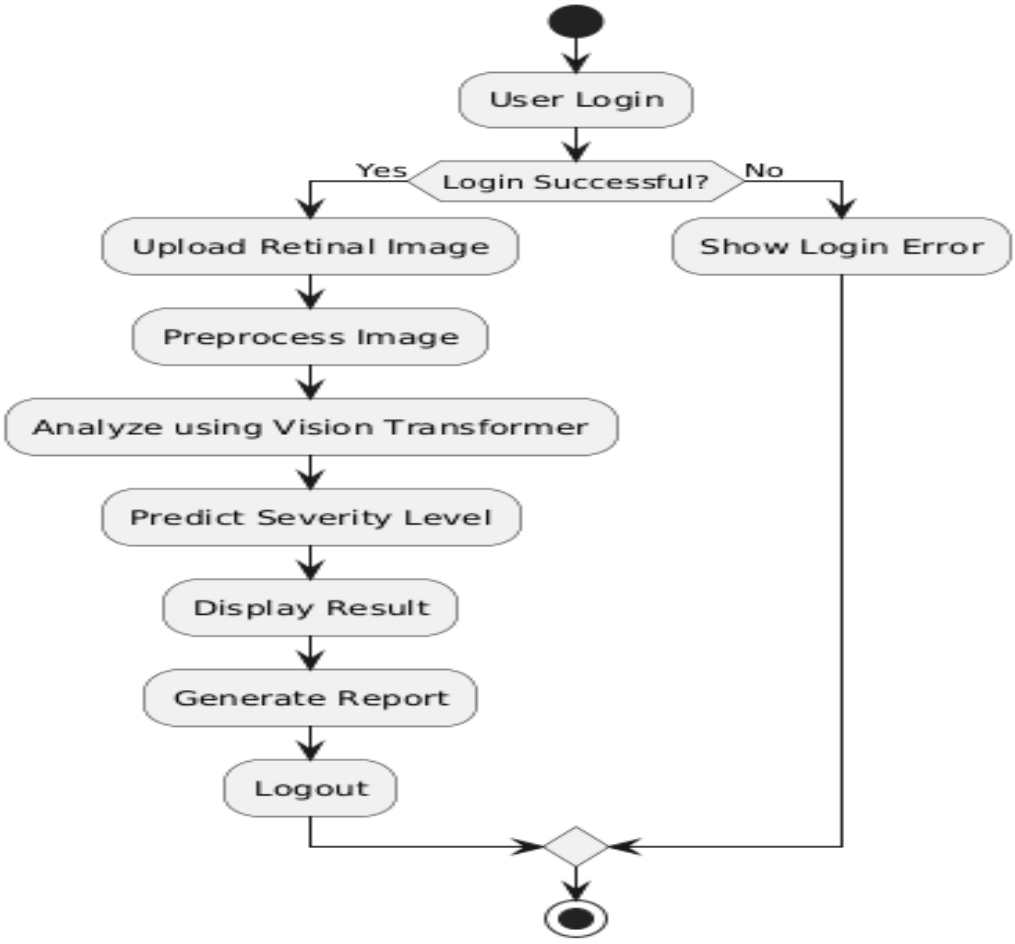


Figure 4.6 Activity Diagram

4.6 Sequence Diagram

A Sequence Diagram represents the interaction between different components of the system over time. It shows how messages are exchanged between objects in the system to perform a specific task. Sequence diagrams are useful for understanding the order in which system operations occur and how different modules communicate with each other during the execution of a process.

In the Diabetic Retinopathy Detection System, the sequence diagram illustrates how the user interacts with the system to upload a retinal image and how the system processes the image using the Vision Transformer model to detect diabetic retinopathy severity. The diagram shows the sequence of operations starting from the user input until the final result is displayed.

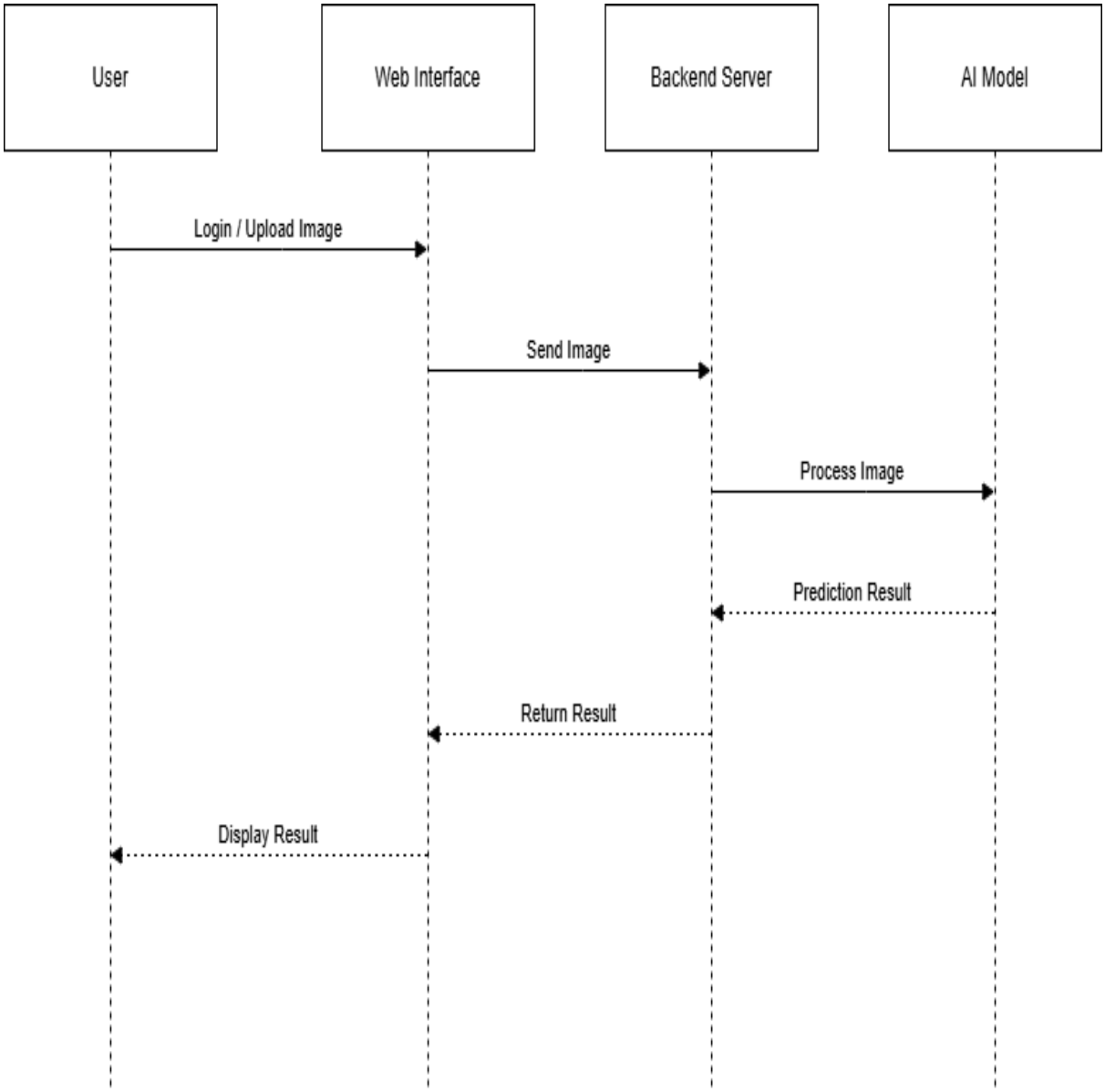


Figure 4.7 Sequence Diagram

• **User** – The person who uploads the retinal image and views the prediction result.

• **User Interface (UI)** – The system interface that receives the retinal image from the user and displays the prediction result.

• **Image Processing Module** – Responsible for validating and preprocessing the uploaded retinal image.

• **Vision Transformer Model** – The deep learning model used to analyze the retinal image and classify the severity of diabetic retinopathy.

• **Result Module** – Generates the prediction result and sends it back to the user interface.

Sequence of Operations

• **Step 1: Upload Retinal Image**

The user uploads a retinal fundus image through the user interface.

• **Step 2: Send Image to Processing Module**

The user interface sends the uploaded image to the image processing module.

• **Step 3: Image Validation and Preprocessing**

The image processing module verifies the image format and performs preprocessing operations such as image resizing, noise removal, contrast enhancement, and normalization.

• **Step 4: Feature Extraction**

The processed image is sent to the Vision Transformer model, which extracts important features from the retinal image.

• **Step 5: Model Prediction**

The Vision Transformer model analyzes the extracted features and predicts the severity level of diabetic retinopathy.

• **Step 6: Generate Prediction Result**

The prediction result along with the confidence score is generated by the result module.

• **Step 7: Return Result to Interface**

The result module sends the classification result back to the user interface.

• **Step 8: Display Output**

The user interface displays the predicted diabetic retinopathy severity level and confidence score to the user.

CHAPTER 5

METHODOLOGY

The methodology describes the approach used to design, implement, and evaluate the RetinaAI: Deep Learning Diagnostic Suite for Diabetic Retinopathy Detection. This system was developed to automatically analyze retinal fundus images and detect the stage of diabetic retinopathy using deep learning techniques.

Diabetic retinopathy is one of the leading causes of vision loss among diabetic patients. Early detection and diagnosis are essential to prevent severe complications. However, manual examination of retinal images by ophthalmologists can be time-consuming and requires specialized expertise. The proposed system addresses this challenge by providing an automated AI-based diagnostic platform that assists medical professionals in identifying diabetic retinopathy stages quickly and accurately.

The system follows a structured pipeline consisting of multiple stages including patient data collection, retinal image upload, image preprocessing, deep learning model analysis, disease classification, report generation, and data storage. The platform is implemented using a web-based interface that allows users to interact with the system easily.

Each component of the system performs a specific task within the diagnostic pipeline to ensure accurate analysis and efficient report generation.

5.1 Module Descriptions

The RetinaAI Diagnostic Suite is composed of several functional modules. Each module performs a specific role in the system workflow. These modules work together to process retinal images, analyze them using deep learning algorithms, and generate diagnostic reports.

The main modules of the system are:

1. Patient Registration Module
2. Retinal Image Upload Module

3. Deep Learning Classification Module
4. Diagnostic Report Generation Module
5. Admin Management Module
6. Patient Diagnostic Archive Module

5.1.1 Patient Registration Module

The Patient Registration Module allows medical staff to register patient details before performing retinal image analysis. This module collects important information such as the patient's name, consultation date, and assigned specialist.

The main purpose of this module is to maintain proper patient records and ensure that diagnostic results are associated with the correct individual. When a patient visits the healthcare facility for retinal screening, the system records the patient's details through a simple input form.

The patient registration interface is implemented using a Streamlit-based web interface, which provides an easy-to-use form for entering patient details. After entering the required information, the system stores the data temporarily until the retinal image is uploaded for analysis.

This module ensures that patient data management is organized and integrated with the diagnostic system.

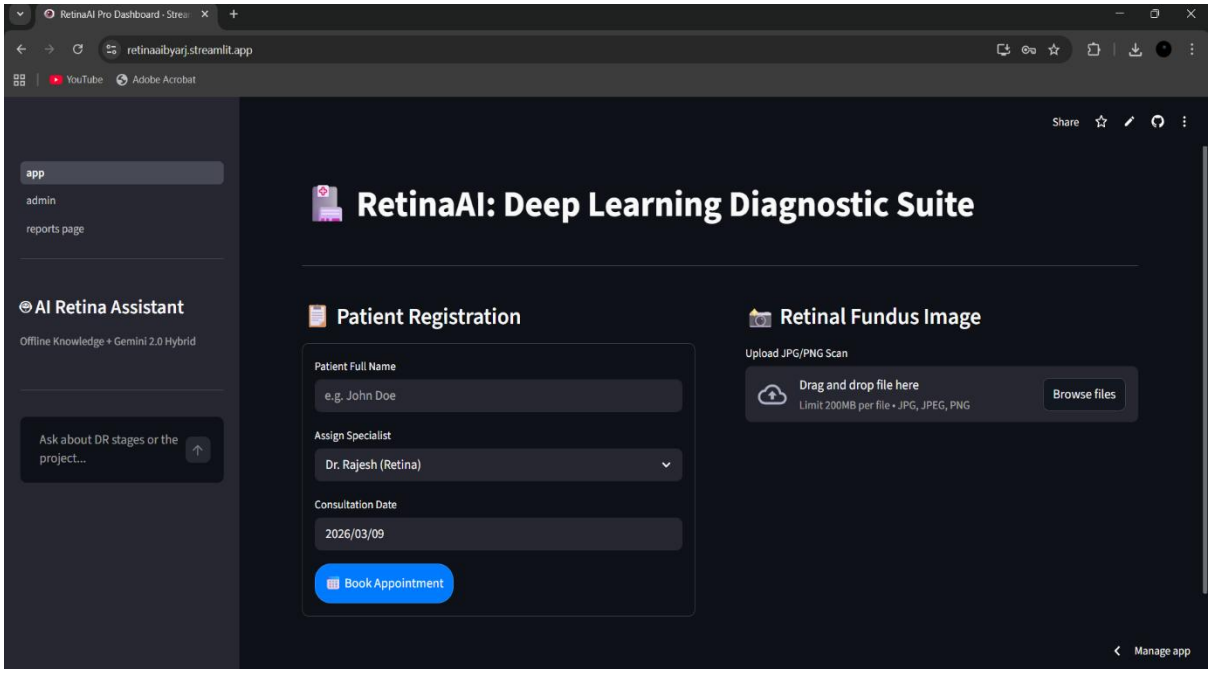


Figure 5.1 Home

5.1.2 Retinal Fundus Image Upload Module

The Retinal Image Upload Module enables users to upload retinal fundus images of patients for automated analysis. These images are typically captured using specialized retinal imaging devices.

Users can upload images in formats such as JPG, JPEG, or PNG. Once the image is uploaded, the system stores the image temporarily for further processing by the deep learning model.

The uploaded retinal image represents the internal structure of the eye, including the blood vessels, optic disc, and macula. These structures contain visual indicators that help identify the severity of diabetic retinopathy.

The upload interface supports drag-and-drop functionality, making it convenient for users to upload retinal scans quickly.

This module acts as the entry point for retinal image analysis within the RetinaAI system.

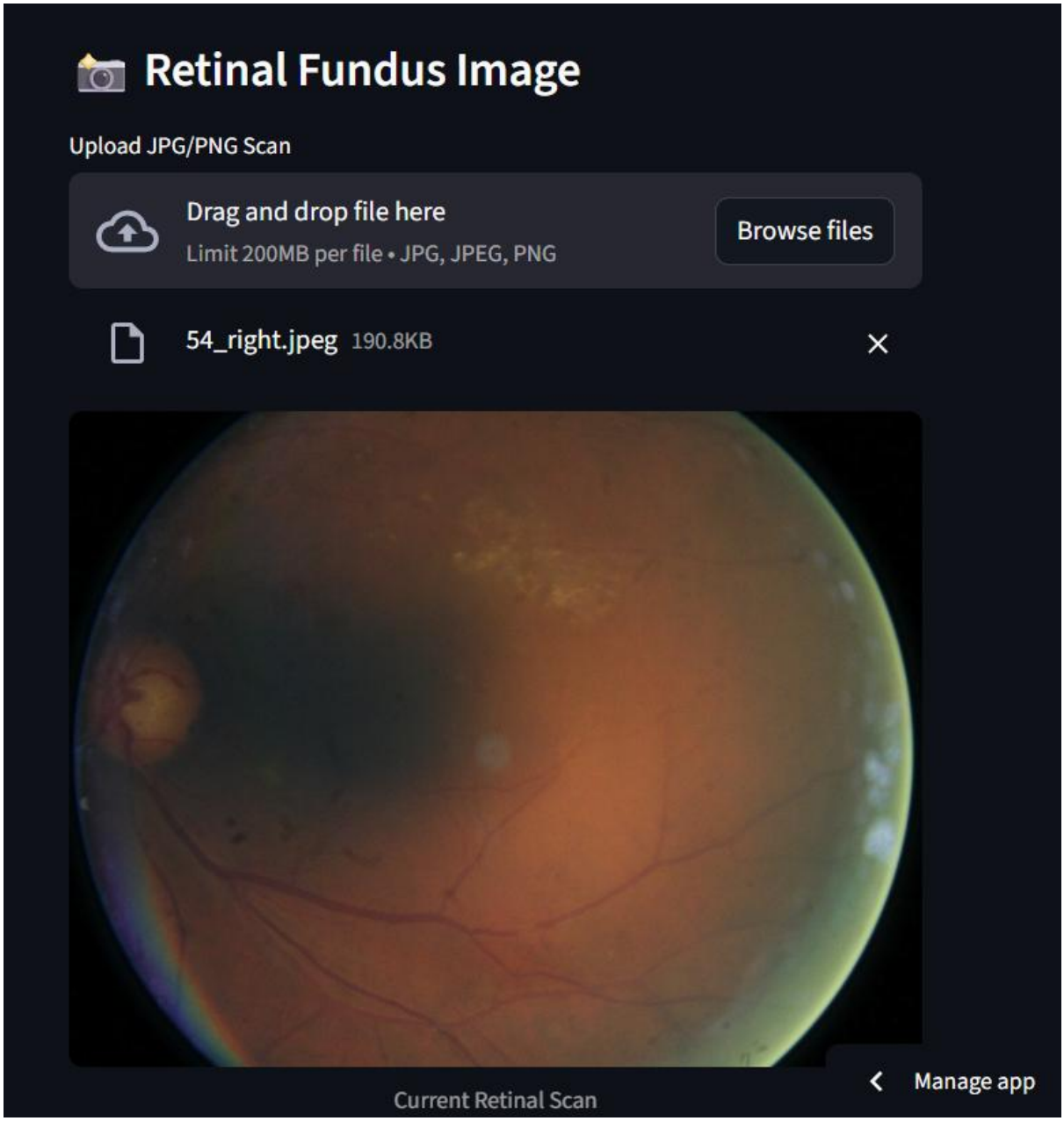


Figure 5.2 Image Upload

5.1.3 Deep Learning Classification Module

The Deep Learning Classification Module is the core component of the RetinaAI system. This module analyzes retinal fundus images using a trained deep learning model to detect diabetic retinopathy.

The system uses a Convolutional Neural Network (CNN) architecture that has been trained on a dataset containing labeled retinal images representing different stages of diabetic retinopathy.

The model extracts visual features such as:

• Microaneurysms

• Hemorrhages

• Exudates

• Blood vessel abnormalities

These features help the system identify the severity of diabetic retinopathy.

The classification model predicts the stage of the disease according to the following categories:

Stage 0 – No Diabetic Retinopathy

Stage 1 – Mild Non-Proliferative Diabetic Retinopathy (NPDR)

Stage 2 – Moderate NPDR

Stage 3 – Severe NPDR

Stage 4 – Proliferative Diabetic Retinopathy (PDR)

The prediction is accompanied by a confidence score that indicates the probability of the predicted stage.

This module plays a critical role in providing automated medical assistance for early disease detection.

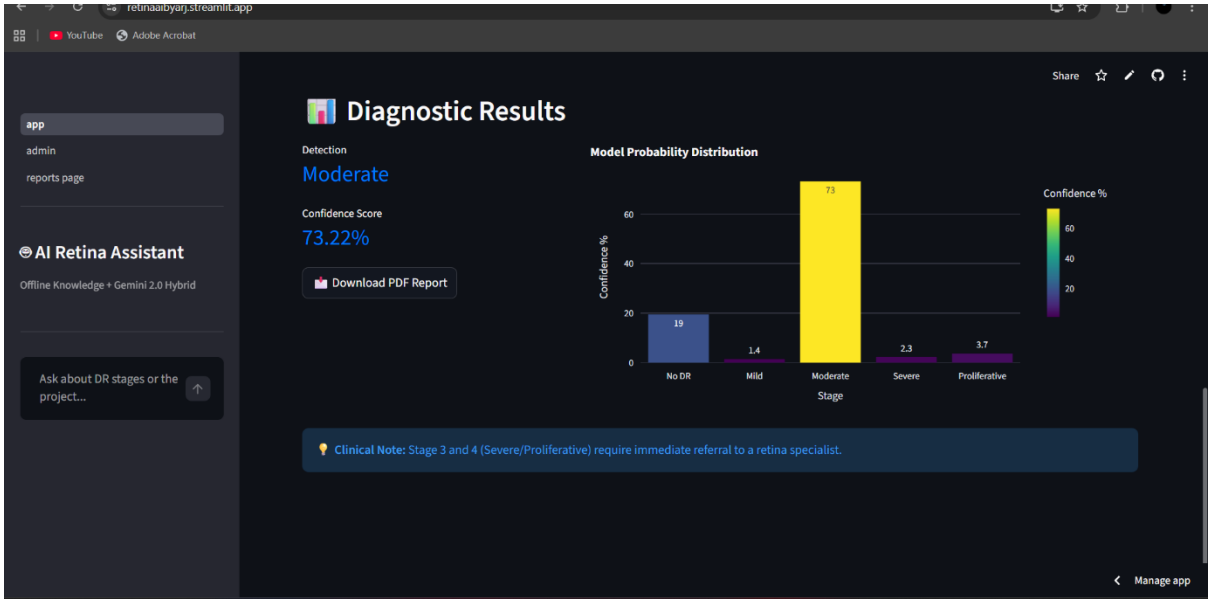


Figure 5.3 Result

5.1.4 Diagnostic Report Generation Module

After the deep learning model generates a prediction, the Diagnostic Report Generation Module creates a structured medical report containing the analysis results.

The report includes the following information:

• Patient name

• Consultation date

• Uploaded retinal image

• Predicted DR stage

• Confidence score of prediction

This report provides a clear summary of the AI-based diagnosis and can be used by doctors to assist in medical decision-making.

The generated reports are displayed through the system interface and stored in the system database for future reference.

RetinaAI Pro Diagnostic Report

Generated on: March 09, 2026 - 12:58

Field	Value
Patient Name:	das
Assigned Doctor:	Dr. Rajesh (Retina)
AI Detection Result:	Moderate
Confidence Score:	73.22%

Analyzed Retinal Fundus Scan:

Figure 5.5 Report Generation

5.1.5 Admin Management Module

The Admin Management Module provides administrative control over the system. This module allows administrators to manage patient records, review diagnostic reports, and monitor system activities.

To access the admin panel, the administrator must enter a secure password through the admin login interface.

The admin dashboard provides features such as:

• Viewing pending patient requests

• Searching patient records

• Accessing diagnostic reports

• Managing patient history

This module ensures that the system maintains proper administrative control and data organization.

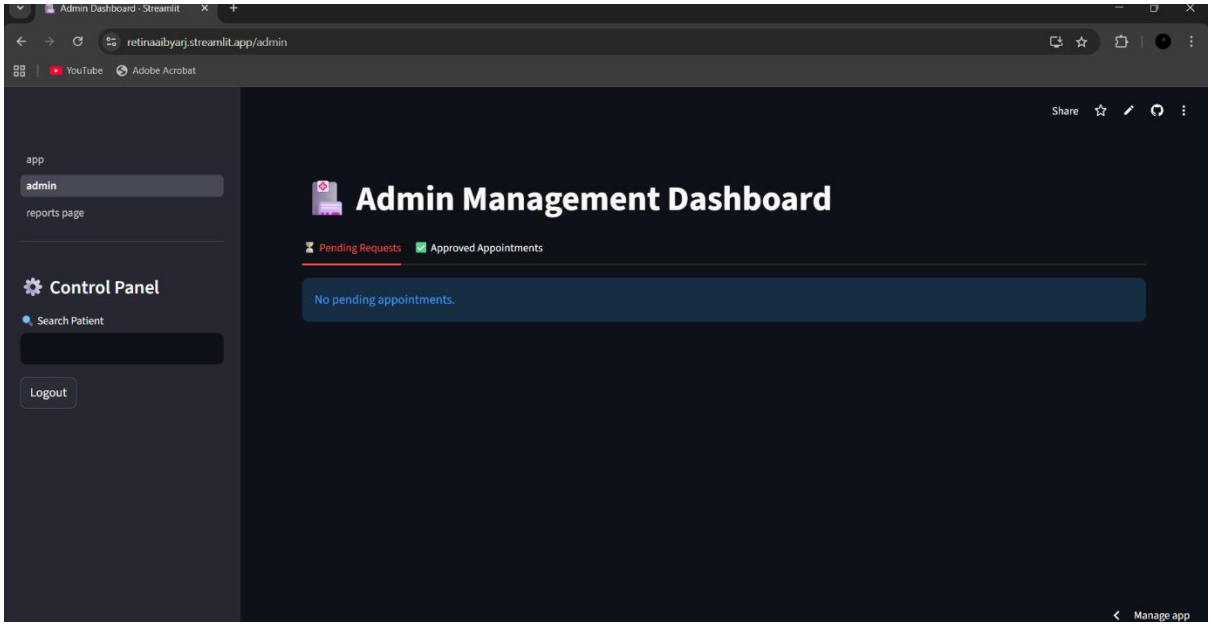


Figure 5.6 Admin Dashboard

5.1.6 Patient Diagnostic Archive Module

The Patient Diagnostic Archive Module stores all previously generated diagnostic reports in the system database.

This module maintains a historical record of patient diagnoses, allowing healthcare professionals to track the progression of diabetic retinopathy over time.

Stored information includes:

• Patient details

• Uploaded retinal images

• Prediction results

• Diagnosis reports

The archive allows doctors to review previous analyses and monitor disease progression during follow-up consultations.

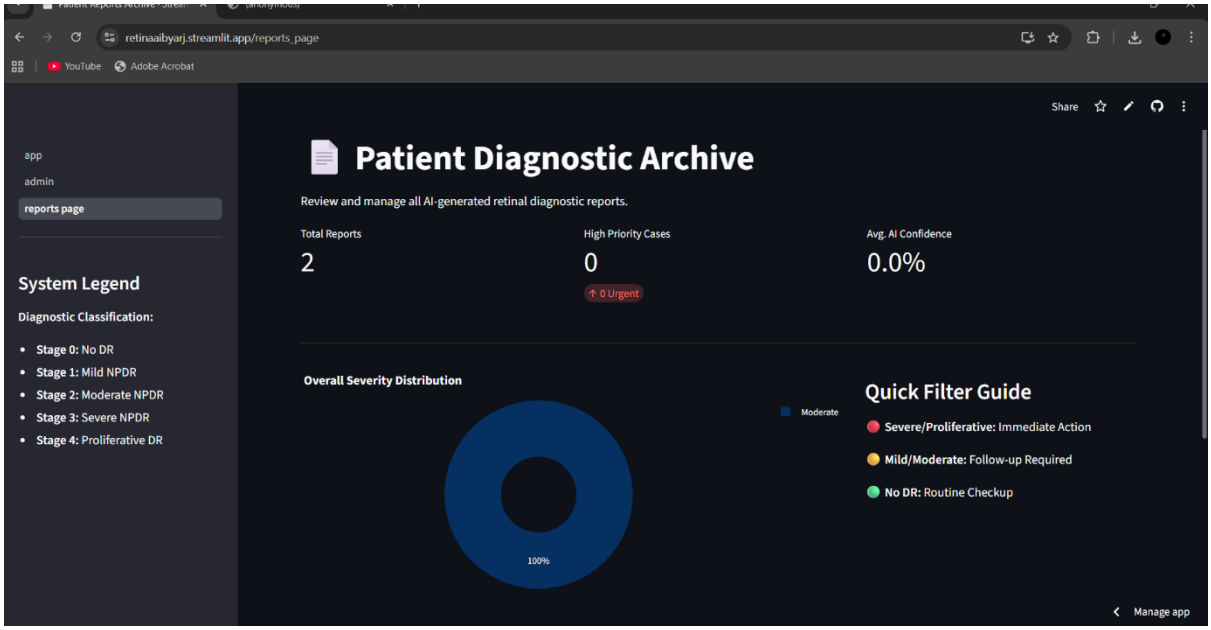


Figure 5.7 Patient Diagnostic Module

5.2 Algorithms

The RetinaAI system uses deep learning algorithms to analyze retinal fundus images and detect the presence and severity of diabetic retinopathy. Algorithms play a critical role in enabling the system to process medical images, extract meaningful visual features, and classify the disease stage automatically. The proposed system primarily relies on image preprocessing techniques and a Convolutional Neural Network (CNN) based classification algorithm to perform accurate retinal image analysis.

The first algorithm used in the system is the image preprocessing algorithm, which prepares the retinal images before they are analyzed by the deep learning model. Retinal fundus images captured by medical imaging devices often contain variations in resolution, lighting conditions, and background noise. These variations may reduce the performance of the deep learning model if the images are not standardized. Therefore, preprocessing is performed to improve image quality and ensure that the input data is consistent. During this process, the retinal image is resized to match the input dimensions required by the neural network model. This ensures that all images are processed with a uniform resolution. After resizing, pixel intensity values are normalized to a smaller numerical range so that the neural network can process the image data efficiently. Noise reduction techniques are also applied to remove unwanted distortions

present in the image. Once preprocessing is completed, the retinal image is converted into a numerical array format that can be interpreted by the deep learning model. This preprocessing algorithm ensures that the retinal images are standardized and suitable for further analysis by the classification model.

The main algorithm used for disease detection in the RetinaAI system is the Convolutional Neural Network (CNN) algorithm. CNN is a specialized deep learning architecture designed for image recognition and computer vision applications. It is particularly effective in medical image analysis because it can automatically learn visual features from images without requiring manual feature extraction. In this system, the CNN model analyzes the retinal fundus image and extracts important visual patterns that indicate the presence of diabetic retinopathy. The CNN model consists of multiple layers that work together to analyze the image. The convolution layers apply filters that scan across the image to detect low-level features such as edges, textures, and shapes. As the image passes through deeper layers of the network, more complex features such as blood vessel abnormalities, hemorrhages, and microaneurysms are identified. Activation functions are applied to introduce non-linear behavior into the model, allowing it to learn complex visual patterns. Pooling layers are used to reduce the spatial size of the feature maps while retaining the most important information. This step helps reduce computational complexity and prevents overfitting.

After feature extraction is completed, the extracted features are passed to fully connected layers that perform the final classification task. These layers analyze the learned features and generate probability scores for different diabetic retinopathy stages. The system then selects the stage with the highest probability as the final prediction result. The CNN model is trained using labeled retinal image datasets that contain examples of different stages of diabetic retinopathy. Through training, the model learns to recognize patterns associated with each disease stage.

The classification results generated by the CNN model correspond to the five standard stages of diabetic retinopathy. These stages include Stage 0 representing no diabetic retinopathy, Stage 1 representing mild non-proliferative diabetic retinopathy, Stage 2 representing moderate non-proliferative diabetic retinopathy, Stage 3 representing severe non-proliferative diabetic retinopathy, and Stage 4 representing proliferative diabetic retinopathy. The system outputs the predicted stage along with a confidence score indicating the probability of the prediction.

By combining image preprocessing techniques with deep learning classification algorithms, the RetinaAI system is capable of analyzing retinal images efficiently and providing accurate diagnostic results. These algorithms enable the automated detection of diabetic retinopathy and assist healthcare professionals in early disease diagnosis..

5.3 System Architecture

The system architecture of the RetinaAI Deep Learning Diagnostic Suite for Diabetic Retinopathy Detection describes the structural design and operational workflow of the proposed system. The architecture is designed to integrate medical image analysis with deep learning techniques in order to automatically detect and classify the severity of diabetic retinopathy from retinal fundus images. The system follows a multi-stage processing pipeline that begins with patient data input and retinal image upload and ends with automated disease classification and diagnostic report generation.

The architecture of the system is composed of several interconnected components that work together to perform retinal image analysis efficiently. The workflow begins when a healthcare professional or medical staff member enters the patient details into the system through the patient registration interface. This information includes patient identification details and consultation data, which are required for maintaining proper medical records. Once the patient information is recorded, the system allows the user to upload a retinal fundus image obtained from a retinal imaging device such as a fundus camera.

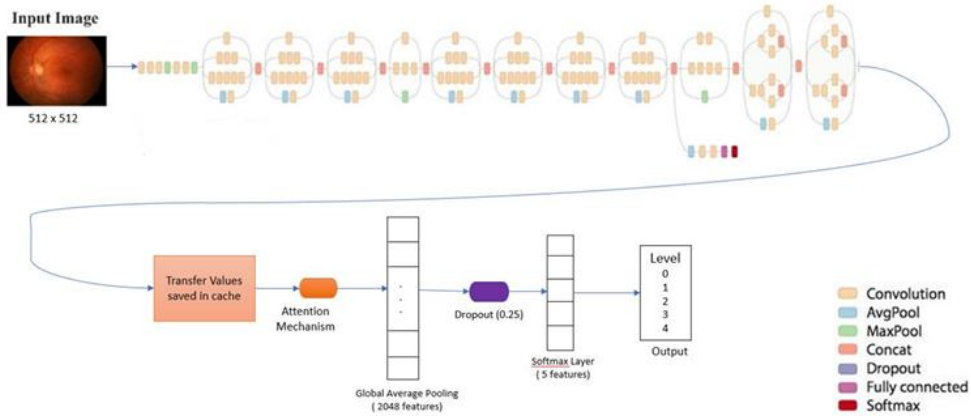


Figure 6: InceptionV3 Architecture (Ref. Milton-Barker, 2019)

Figure 5.8 Architecture

After the retinal image is uploaded, the system passes the image to the image preprocessing module. The preprocessing stage prepares the retinal image for analysis by performing operations such as resizing the image, normalizing pixel values, and reducing image noise. These preprocessing operations ensure that the retinal images are standardized and compatible with the deep learning model. By improving image quality and consistency, the preprocessing module enhances the accuracy and reliability of the classification process.

Once preprocessing is completed, the processed retinal image is forwarded to the deep learning analysis module. This module contains the trained Convolutional Neural Network (CNN) model that performs the core image analysis. The CNN model processes the retinal image through multiple convolution layers that extract important visual features from the image. These features include patterns associated with diabetic retinopathy such as microaneurysms, hemorrhages, exudates, and abnormalities in retinal blood vessels. As the image passes through deeper layers of the neural network, the system learns complex patterns that are characteristic of different stages of diabetic retinopathy.

After feature extraction, the CNN model performs the classification process. The classification layer analyzes the extracted features and determines the severity stage of diabetic retinopathy present in the retinal image. The system classifies the retinal image into one of five categories representing different stages of the disease. These categories include no diabetic retinopathy, mild non-proliferative diabetic retinopathy, moderate non-proliferative diabetic retinopathy,

severe non-proliferative diabetic retinopathy, and proliferative diabetic retinopathy. The system also calculates a confidence score that indicates the probability of the predicted stage.

Following the classification process, the prediction results are passed to the diagnostic report generation module. This module generates a structured report that contains patient details, retinal image information, predicted disease stage, and the confidence score produced by the deep learning model. The report provides a clear summary of the automated diagnosis and can be used by doctors to support clinical decision-making. The generated report is displayed to the user through the system interface and can also be downloaded or saved for medical documentation purposes.

The final component of the system architecture is the data storage and administration module. In this stage, all patient records, uploaded retinal images, and diagnostic reports are stored in the patient diagnostic archive. This archive maintains a historical database of patient diagnoses and allows healthcare professionals to review previous reports and monitor disease progression over time. The system also includes an administrative dashboard that allows authorized users to manage patient records, search diagnostic reports, and monitor system activity.

Overall, the RetinaAI system architecture integrates deep learning based retinal image analysis with a web-based medical management system. By combining patient data management, automated image classification, and diagnostic report generation, the system provides an efficient and reliable platform for detecting diabetic retinopathy. The architecture supports early disease detection and assists healthcare professionals in diagnosing retinal diseases more effectively, thereby contributing to improved patient care and prevention of vision loss.

CHAPTER 6

IMPLEMENTATION DETAILS

The implementation phase involves converting the system design and proposed methodology into a functional software application. In this project, an automated system was developed to detect and classify the severity of diabetic retinopathy using retinal fundus images. The system integrates deep learning techniques, image preprocessing methods, and a Vision Transformer model to analyze retinal images and predict the stage of diabetic retinopathy.

The implementation focuses on developing a complete pipeline that allows users to upload retinal images, preprocess the images to improve quality, extract relevant features using deep learning models, classify the disease severity, and display the results through an easy-to-use interface.

The implementation process was divided into several stages including environment setup, dataset preparation, image preprocessing, model development, model training, and user interface implementation.

6.1 Implementation Procedures

The development of the diabetic retinopathy detection system was carried out in several stages. Each stage focuses on a specific component of the system.

The major implementation stages include:

• Development environment setup

• Dataset collection and preparation

• Image preprocessing

• Feature extraction

• Vision Transformer model implementation

• Model training and evaluation

• User interface development

• Result visualization

6.1.1 Development Environment

The development environment includes the hardware and software tools required to build and run the diabetic retinopathy detection system.

The project was implemented using Python, which is widely used in machine learning and deep learning applications due to its simplicity and powerful libraries.

Programming Language

Python was selected as the primary programming language because it provides extensive support for machine learning, image processing, and deep learning tasks**.**

Framework

The system interface was developed using Streamlit or Flask, which allows developers to build interactive web applications using Python.

Libraries Used

Several Python libraries were used in the implementation of this system.

NumPy

NumPy provides support for numerical computations and array operations. It is widely used for handling image data and mathematical calculations.

Pandas

Pandas is used for data manipulation and dataset handling.

OpenCV

OpenCV is used for image processing tasks such as image resizing, filtering, and noise removal.

Pillow

Pillow is used for image loading and image format handling.

TensorFlow / PyTorch

These deep learning frameworks were used to implement the Vision Transformer model.

Matplotlib and Seaborn

These libraries were used to visualize training results and generate graphs.

Scikit-learn

Scikit-learn was used for evaluation metrics and data preprocessing.

6.1.2 Dataset Collection

The performance of the diabetic retinopathy detection system depends largely on the quality and size of the dataset used for training.

For this project, retinal fundus images were collected from publicly available datasets used in medical research.

EyePACS Dataset

The EyePACS dataset is one of the most commonly used datasets for diabetic retinopathy detection. It contains thousands of retinal images labeled according to diabetic retinopathy severity levels.

Messidor Dataset

The Messidor dataset contains high-resolution retinal images along with medical annotations indicating the severity level of diabetic retinopathy.

Each retinal image in these datasets is labeled into one of the following classes:

- **No Diabetic Retinopathy**
- **Mild Diabetic Retinopathy**
- **Moderate Diabetic Retinopathy**
- **Severe Diabetic Retinopathy**
- **Proliferative Diabetic Retinopathy**

6.1.3 Image Preprocessing Implementation

Before training the deep learning model, retinal images must undergo preprocessing to improve image quality and remove unwanted noise.

Raw retinal images may contain variations in brightness, contrast, and resolution. These factors can negatively affect model performance.

The following preprocessing techniques were applied:

Image Resizing

All retinal images were resized to a fixed resolution so that they could be processed efficiently by the deep learning model.

Noise Removal

Noise in the retinal images was removed using filtering techniques to improve image clarity.

Contrast Enhancement

Contrast enhancement was applied to highlight retinal structures such as blood vessels and lesions.

Normalization

Normalization was applied to scale pixel values into a consistent range, which helps improve model convergence during training.

6.1.4 Feature Extraction Implementation

Deep learning models require numerical representations of image features in order to perform classification.

In this project, the Vision Transformer model was used to automatically extract important features from retinal images.

The Vision Transformer divides each image into smaller patches. Each patch is converted into an embedding vector that represents visual information.

These embeddings are then processed through transformer layers that capture relationships between different image regions.

Feature extraction allows the model to identify patterns associated with diabetic retinopathy such as:

• Microaneurysms

• Hemorrhages

• Exudates

• Blood vessel abnormalities

6.1.5 Vision Transformer Model Implementation

The Vision Transformer (ViT) model plays a crucial role in the diabetic retinopathy detection system. It is responsible for analyzing retinal fundus images and identifying patterns associated with diabetic retinopathy. Unlike traditional convolutional neural networks, the Vision Transformer uses a transformer architecture that processes images by dividing them into

smaller patches and analyzing the relationships between these patches using attention mechanisms.

In the implementation of the Vision Transformer model, each retinal image is first divided into multiple fixed-size patches. These patches are then converted into vector representations called patch embeddings. Positional encoding is added to these embeddings to preserve spatial information about the location of each patch within the image.

The patch embeddings are passed through a series of transformer encoder layers. Each encoder layer consists of multi-head self-attention mechanisms and feed-forward neural networks. The self-attention mechanism allows the model to focus on important regions of the retinal image and identify features related to diabetic retinopathy such as microaneurysms, hemorrhages, and exudates.

After passing through the transformer layers, the extracted features are sent to a classification layer. This layer predicts the severity level of diabetic retinopathy based on the learned image features. The output of the model corresponds to one of the predefined diabetic retinopathy classes.

The Vision Transformer architecture provides improved performance for image classification tasks because it captures both local and global image features. This capability makes it suitable for analyzing complex retinal images where disease patterns may appear in different regions of the retina.

6.1.6 Web Interface Implementation

The user interface of the system was implemented using Streamlit. Streamlit allows developers to create web applications using Python without requiring complex frontend development. It provides a simple and interactive interface that enables users to upload retinal images and obtain prediction results easily.

The interface includes several components.

Input Section

Users can submit retinal images using:

• Image upload option

• Retinal fundus image selection from local system

This section allows users to upload retinal images obtained from medical imaging devices such as fundus cameras. The uploaded image serves as the primary input for the diabetic retinopathy detection system.

Prediction Section

After the retinal image is uploaded, the system processes the image using preprocessing techniques and the Vision Transformer model. The prediction section displays the classification result indicating the severity level of diabetic retinopathy.

The predicted output may include one of the following classes:

• No Diabetic Retinopathy

• Mild Diabetic Retinopathy

• Moderate Diabetic Retinopathy

• Severe Diabetic Retinopathy

• Proliferative Diabetic Retinopathy

Visualization Section

The interface displays visual representations of the analysis results such as:

• Retinal image preview

• Highlighted abnormal regions

• Prediction confidence graph

These visualizations help users understand how the model analyzes retinal images and detects diabetic retinopathy.

The interactive interface improves user experience and makes the system easier to use for medical professionals and researchers.

6.2 Testing Methods and Results

Testing is an important step in the implementation process. It ensures that the system functions correctly and produces accurate predictions. The diabetic retinopathy detection system was tested using several evaluation techniques to verify the performance of the deep learning model and the overall system functionality.

Testing helps identify potential errors and ensures that the system performs reliably under different conditions.

6.2.1 Dataset Testing

The dataset used for diabetic retinopathy detection was divided into two parts:

• Training Dataset

Used to train the Vision Transformer model.

• Testing Dataset

Used to evaluate the performance of the trained model.

A commonly used approach is the 80–20 split, where the dataset is divided as follows:

• 80% training data

• 20% testing data

The training dataset helps the model learn patterns associated with diabetic retinopathy, while the testing dataset is used to evaluate how well the model performs on unseen retinal images.

This approach ensures that the model generalizes well and does not simply memorize the training data.

6.2.2 Model Evaluation Metrics

Several evaluation metrics were used to measure the performance of the diabetic retinopathy detection model.

Accuracy

Accuracy measures the percentage of correctly classified retinal images.

Accuracy = (Correct Predictions / Total Predictions) × 100

A higher accuracy value indicates better performance of the classification model.

Precision

Precision measures how many predicted diabetic retinopathy cases are actually correct.

Precision = True Positives / (True Positives + False Positives)

Precision is important in medical diagnosis systems because incorrect predictions may lead to unnecessary treatment.

Recall

Recall measures how many actual diabetic retinopathy cases were correctly identified by the model.

Recall = True Positives / (True Positives + False Negatives)

High recall ensures that the system can detect most cases of diabetic retinopathy.

F1 Score

The F1 score combines precision and recall into a single evaluation metric.

F1 Score = 2 × (Precision × Recall) / (Precision + Recall)

This metric provides a balanced evaluation of the classification model.

6.2.3 Confusion Matrix

A confusion matrix was used to evaluate the classification results of the diabetic retinopathy detection model.

The confusion matrix contains four main components:

• True Positive (TP) – Diabetic retinopathy correctly detected

• True Negative (TN) – Healthy retinal image correctly classified

• False Positive (FP) – Healthy retinal image incorrectly classified as diabetic retinopathy

• False Negative (FN) – Diabetic retinopathy incorrectly classified as healthy

The confusion matrix provides a clear visualization of the classification performance of the model and helps identify potential classification errors.

6.2.4 System Testing

In addition to model evaluation, the entire system was tested to ensure proper functionality.

Input Validation Testing

The system verifies whether the uploaded retinal image is valid and in the correct format before processing it.

Prediction Testing

The system verifies that the Vision Transformer model generates predictions correctly for different retinal images.

Visualization Testing

The system ensures that the graphical outputs such as prediction confidence graphs and retinal image previews are displayed correctly.

Interface Testing

The system verifies that the web interface operates correctly and allows users to upload images and view results without errors.

6.2.5 Testing Results

The experimental results showed that the diabetic retinopathy detection model achieved high classification accuracy when tested on unseen retinal images.

The Vision Transformer model successfully identified retinal abnormalities and classified images into different diabetic retinopathy severity levels.

The results demonstrate that the proposed system can effectively assist medical professionals in detecting diabetic retinopathy and can be used for automated retinal image analysis.

CHAPTER 7

ADVANTAGES AND DISADVANTAGES

The Diabetic Retinopathy Detection System was designed to automatically analyze retinal fundus images and determine the severity level of diabetic retinopathy using deep learning techniques and medical image analysis. The system aims to assist healthcare professionals in detecting diabetic retinopathy at an early stage and reducing the risk of vision loss caused by delayed diagnosis.

While the proposed system provides several benefits in improving medical diagnosis and retinal image analysis, it also has certain limitations due to challenges in medical imaging, dataset availability, and computational requirements.

This chapter discusses the advantages and disadvantages of the proposed system to provide a balanced evaluation of its capabilities and performance.

7.1 Advantages

The Diabetic Retinopathy Detection System offers several advantages in terms of accuracy, efficiency, and usability. The system provides an automated approach for analyzing retinal images and assists healthcare professionals in detecting diabetic retinopathy quickly and effectively.

- **Automated Diabetic Retinopathy Detection**

One of the major advantages of the proposed system is its ability to automatically detect diabetic retinopathy from retinal images. Traditional diagnosis methods require ophthalmologists to manually examine retinal images and identify disease-related abnormalities. The automated system uses deep learning algorithms to analyze retinal fundus images and detect patterns associated with diabetic retinopathy. This automated approach reduces the dependency on manual examination and allows faster diagnosis.

By automating the detection process, the system can assist doctors in screening large numbers of patients efficiently.

- **Faster Diagnosis Process**

Manual analysis of retinal images can be time-consuming and requires significant expertise. The proposed system can analyze retinal images quickly and provide classification results within a few seconds.

The rapid diagnosis capability of the system is particularly useful in large hospitals and screening programs where thousands of patients require retinal examination.

Faster analysis helps healthcare professionals make quicker medical decisions and begin treatment at an early stage**.**

- **Advanced Deep Learning Analysis**

The proposed system uses **Vision Transformer architecture**, which is a modern deep learning technique used for image analysis. Vision Transformers analyze images by dividing them into smaller patches and processing them using attention mechanisms.

This approach allows the system to capture both local and global image features. As a result, the model can detect subtle abnormalities in retinal images that may not be easily visible during manual examination.

The use of advanced deep learning models improves the accuracy and reliability of diabetic retinopathy detection.

- **Visualization of Prediction Results**

Another important advantage of the system is the ability to provide visual analysis of prediction results. Instead of displaying only classification results, the system can present additional visual information to help users understand the model predictions.

Visual outputs may include highlighted retinal regions, prediction confidence graphs, and image previews.

These visual representations help medical professionals understand how the model analyzes retinal images and improve trust in the system.

- **User-Friendly Interface**

The system includes a web-based interface developed using Streamlit, which provides a simple and interactive platform for users.

Users can easily upload retinal images and obtain prediction results without requiring technical knowledge of machine learning or programming.

The interface displays the uploaded retinal image, classification result, and confidence score in an organized format. This makes the system accessible for doctors, researchers, and medical staff.

- **Scalable System Architecture**

The architecture of the system is designed in a modular manner. This means that different components such as the preprocessing module, deep learning model, and user interface can be modified or upgraded independently.

For example, the Vision Transformer model can be replaced with more advanced deep learning architectures in the future without redesigning the entire system.

This scalability allows the system to evolve as new technologies and improved models become available.

- **Support for Large-Scale Screening**

Automated diabetic retinopathy detection systems are particularly useful for large-scale screening programs.

Hospitals and healthcare organizations can use the system to analyze large numbers of retinal images and identify patients who may require further medical examination.

This capability can help reduce the burden on ophthalmologists and improve early detection of diabetic retinopathy.

7.2 Disadvantages

Despite its advantages, the Diabetic Retinopathy Detection System also has certain limitations. These limitations mainly arise from the complexity of medical image analysis, variations in retinal image quality, and the dependency of deep learning models on large and high-quality datasets.

Understanding these limitations is important because it helps researchers identify the challenges involved in automated diabetic retinopathy detection and highlights the need for continuous improvement in medical image analysis systems.

- **Dependence on Dataset Quality**

The performance of the diabetic retinopathy detection model depends heavily on the quality and diversity of the dataset used for training. Deep learning models require large amounts of high-quality labeled retinal images in order to learn meaningful patterns associated with different stages of diabetic retinopathy.

If the dataset contains low-resolution images, incorrect labels, or insufficient examples of certain disease stages, the model may not learn accurate features. For example, if the dataset mainly contains images of early-stage diabetic retinopathy, the model may struggle to accurately identify severe or proliferative stages of the disease.

Additionally, datasets collected from different hospitals may vary in terms of lighting conditions, camera types, and image resolution. These variations may reduce the model's ability to generalize well when analyzing new retinal images.

Therefore, the availability of large and well-annotated retinal image datasets is essential for achieving reliable performance.

- **Difficulty in Detecting Subtle Retinal Abnormalities**

Diabetic retinopathy often begins with very small changes in the retina, such as microaneurysms and tiny hemorrhages. These abnormalities may be extremely difficult to detect, even with advanced deep learning techniques.

Although the Vision Transformer model can analyze complex retinal patterns, certain subtle features may still be difficult for automated systems to identify accurately. In some cases, the model may overlook very small lesions or misinterpret certain image patterns.

This limitation highlights the importance of using automated systems as decision-support tools rather than replacing medical experts. Ophthalmologists still play a critical role in verifying diagnosis results and providing medical treatment.

- **Limited Generalization Across Different Imaging Devices**

Another limitation of the proposed system is the variation in retinal images captured using different imaging devices. Retinal fundus images obtained from different cameras may differ in resolution, brightness, color balance, and image quality.

Deep learning models trained on a specific dataset may perform well on similar images but may not perform equally well on images captured using different equipment or imaging conditions.

For example, retinal images captured in hospitals with advanced medical equipment may differ significantly from images captured in smaller clinics with lower-resolution cameras.

This variability can affect the prediction accuracy of the model when analyzing images from different sources.

- **Dependence on Image Quality**

The accuracy of the diabetic retinopathy detection system is highly dependent on the quality of the uploaded retinal images. Images that are blurred, poorly illuminated, or improperly captured may contain insufficient visual information for accurate analysis.

Poor image quality may occur due to factors such as improper camera focus, patient eye movement, inadequate lighting, or technical limitations of imaging devices.

When low-quality images are provided as input, the system may produce incorrect predictions or fail to detect important retinal abnormalities.

Therefore, ensuring high-quality retinal image acquisition is essential for achieving reliable diagnostic results.

- **Limited Clinical Context Understanding**

The proposed system primarily analyzes retinal images to detect diabetic retinopathy. However, medical diagnosis often requires additional clinical information such as patient history, blood glucose levels, blood pressure, and other medical records.

Without incorporating this additional clinical information, the system may not provide a complete understanding of the patient's overall health condition.

For example, two patients may have similar retinal images but different medical histories, which may influence the diagnosis and treatment decisions.

Future systems should integrate clinical data with retinal image analysis to improve diagnostic accuracy.

- **Computational Resource Requirements**

Deep learning models such as Vision Transformers require significant computational resources for training and optimization. Training these models on large retinal image datasets often requires high-performance GPUs and large memory capacity.

The computational requirements increase further when training more advanced models or processing large datasets.

Although the trained model can perform predictions relatively quickly, the training process itself can be time-consuming and resource-intensive.

This may pose challenges for institutions with limited computational infrastructure.

Summary

Although the proposed Diabetic Retinopathy Detection System offers several benefits in terms of automated diagnosis and faster retinal image analysis, it also faces certain limitations related

to dataset availability, image quality, computational requirements, and medical context understanding.

Addressing these challenges requires continued research, larger and more diverse datasets, improved imaging techniques, and integration of additional medical information.

By overcoming these limitations, future diabetic retinopathy detection systems can become more accurate, reliable, and useful in real-world healthcare environments.

CHAPTER 8

RESULT AND DISCUSSION

The Results and Discussion chapter presents the outcomes obtained from the implementation of the Diabetic Retinopathy Detection System. This chapter evaluates how effectively the system performs retinal image analysis using deep learning techniques and Vision Transformer architecture. It also discusses the insights obtained from the classification results and visualization outputs generated by the system.

The primary objective of this project is to develop a system capable of identifying the severity level of diabetic retinopathy from retinal fundus images using automated analysis. The system processes retinal images through several stages including image preprocessing, feature extraction, deep learning classification using Vision Transformer models, and visualization of prediction results.

The experimental results demonstrate the ability of the system to analyze retinal images, identify disease-related abnormalities, and provide meaningful predictions regarding the severity level of diabetic retinopathy. The results also show that the use of advanced deep learning techniques improves the accuracy and reliability of automated diabetic retinopathy detection.

8.1 System Output Overview

When a user uploads a retinal fundus image into the platform, the system performs a complete analysis and produces several outputs. These outputs help users understand the condition of the retina and determine whether diabetic retinopathy is present.

The main outputs generated by the system include:

• Diabetic retinopathy severity prediction

• Confidence score of the prediction

• Retinal image visualization

• Detected abnormal regions in the retina

• Graphical visualization of prediction results

The system interface presents these outputs through an interactive dashboard that displays the uploaded retinal image, classification results, and graphical insights. This interface allows medical professionals and researchers to easily interpret the results generated by the system.

8.2 Prediction Results

The diabetic retinopathy detection model classifies retinal images into different disease severity categories. These categories help doctors and medical professionals understand the progression of the disease and determine appropriate treatment strategies.

The possible prediction outcomes generated by the system include:

No Diabetic Retinopathy

This classification indicates that the retinal image does not show any visible signs of diabetic retinopathy. The retinal blood vessels appear normal and no abnormal lesions are detected.

Mild Diabetic Retinopathy

This stage represents the early stage of diabetic retinopathy where small abnormalities such as microaneurysms appear in the retina. These small bulges in retinal blood vessels may not immediately affect vision but require monitoring.

Moderate Diabetic Retinopathy

In this stage, the disease begins to progress and more retinal blood vessels become damaged. Hemorrhages and exudates may appear in the retinal image, indicating worsening retinal damage.

Severe Diabetic Retinopathy

At this stage, many retinal blood vessels are blocked, reducing blood supply to the retina. The retina may begin to signal the growth of new abnormal blood vessels.

Proliferative Diabetic Retinopathy

This is the most advanced stage of the disease. New abnormal blood vessels grow in the retina and may leak blood into the vitreous fluid of the eye. This condition can lead to severe vision loss if not treated promptly.

For example, when the system analyzed a retinal image, the following output was generated:

Prediction Result: Moderate Diabetic Retinopathy

Confidence Score: 92.4%

This indicates that the retinal image shows visible signs of diabetic retinopathy at the moderate stage and requires medical attention.

8.3 Feature Detection and Abnormality Identification

One of the important capabilities of the proposed system is its ability to identify abnormal features present in retinal images. The Vision Transformer model analyzes the retinal image and detects important features associated with diabetic retinopathy.

Some of the common retinal abnormalities detected by the system include:

• Microaneurysms

• Hemorrhages

• Hard exudates

• Cotton wool spots

• Blood vessel abnormalities

These features are important indicators used by ophthalmologists to diagnose diabetic retinopathy. By automatically detecting these abnormalities, the system assists medical professionals in identifying the severity level of the disease.

The detection of multiple abnormal features increases the likelihood that the retinal image belongs to a higher severity stage of diabetic retinopathy.

8.4 Disease Severity Index

The system also generates a disease severity score that represents the probability of diabetic retinopathy in the analyzed retinal image. This score is calculated based on the prediction confidence generated by the deep learning model.

The severity score ranges from 0 to 100**,** where higher values indicate a higher likelihood of diabetic retinopathy.

For example:

Severity Score: **92.4%**

This score indicates that the retinal image contains strong evidence of diabetic retinopathy and requires further medical evaluation.

The severity score is displayed using graphical visualizations that allow users to easily interpret the prediction results.

8.5 Retinal Feature Visualization

The system also provides visual representations of the analyzed retinal images. These visualizations help users understand how the model identifies abnormalities within the retinal image.

The system may highlight regions of interest within the retinal image where abnormal patterns are detected. These highlighted regions correspond to areas where the model identifies potential lesions or blood vessel abnormalities.

Feature visualization helps improve the interpretability of the model and allows doctors to verify whether the system's predictions correspond to visible retinal abnormalities.

CHAPTER 9

CONCLUSION

9.1 Summary

Diabetic retinopathy is one of the leading causes of vision impairment and blindness among people with diabetes. Early detection and timely treatment are essential to prevent severe vision loss. However, manual analysis of retinal fundus images by ophthalmologists can be time-consuming and requires specialized medical expertise. With the advancement of artificial intelligence and deep learning technologies, automated systems have become an effective solution for assisting medical professionals in diagnosing retinal diseases.

This project aimed to develop an intelligent system capable of detecting and classifying diabetic retinopathy from retinal fundus images using deep learning techniques. The proposed system, RetinaAI – Deep Learning Diagnostic Suite for Diabetic Retinopathy Detection, provides an automated platform that analyzes retinal images and predicts the severity stage of diabetic retinopathy. The system integrates image processing techniques, deep learning algorithms, and a web-based interface to create an efficient diagnostic support tool.

The proposed system follows a structured processing pipeline consisting of multiple stages including patient data input, retinal image upload, image preprocessing, deep learning based classification, diagnostic report generation, and visualization of results. During the analysis process, retinal fundus images are processed using preprocessing techniques to standardize image size, normalize pixel values, and reduce noise. These preprocessing operations improve the quality of the images and enhance the performance of the deep learning model.

The system uses a Convolutional Neural Network (CNN) based deep learning model to analyze retinal images and extract meaningful visual features related to diabetic retinopathy. The model learns patterns associated with retinal abnormalities such as microaneurysms, hemorrhages, exudates, and blood vessel irregularities. Based on these extracted features, the model classifies the retinal image into different stages of diabetic retinopathy, including no diabetic retinopathy, mild non-proliferative diabetic retinopathy, moderate non-proliferative diabetic retinopathy, severe non-proliferative diabetic retinopathy, and proliferative diabetic retinopathy.

Another important feature of the system is the automated diagnostic report generation module. After analyzing the retinal image, the system generates a detailed report containing patient information, the uploaded retinal image, predicted disease stage, and prediction confidence score. This report can assist doctors in making informed clinical decisions and maintaining proper patient records.

The system also includes an administrative management interface that allows authorized users to manage patient records, review diagnostic reports, and monitor system activity. In addition, the patient diagnostic archive stores historical diagnostic reports, enabling healthcare professionals to track disease progression over time.

The implementation results demonstrate that the proposed RetinaAI system is capable of analyzing retinal images efficiently and generating diagnostic results within a short period of time. The use of deep learning techniques significantly improves the accuracy and reliability of diabetic retinopathy detection. Overall, the system provides an effective tool that can assist healthcare professionals in early disease detection and improve the efficiency of retinal screening programs.

9.2 Limitations

Although the proposed RetinaAI system provides an effective approach for automated diabetic retinopathy detection, certain limitations still exist. Medical image analysis is a complex task, and the accuracy of automated diagnostic systems can be influenced by various factors.

One limitation of the system is its dependence on the dataset used for training the deep learning model. The performance of the model depends on the diversity and quality of the training data. If the dataset does not contain sufficient examples of different retinal conditions, the model may have difficulty accurately classifying unseen images.

Another limitation is related to image quality. Retinal images captured under poor lighting conditions or with low-resolution imaging devices may contain noise or distortions that can affect classification accuracy. In such cases, the system may struggle to detect subtle retinal abnormalities.

Additionally, the current system primarily focuses on analyzing retinal images for diabetic retinopathy detection. However, many other retinal diseases such as glaucoma, macular degeneration, and retinal detachment may also be present in retinal images. Since the system is specifically trained for diabetic retinopathy detection, it may not identify other eye diseases.

The system also relies on automated predictions generated by the deep learning model. Although the model provides high accuracy, it should not replace professional medical diagnosis. Instead, it should be used as a decision-support tool to assist ophthalmologists in the diagnostic process.

9.3 Future Scope

There are several opportunities to further improve and extend the RetinaAI system in the future. Advances in artificial intelligence and medical imaging technologies provide many possibilities for enhancing automated retinal disease detection systems.

One potential improvement is the integration of more advanced deep learning architectures such as Vision Transformers (ViT) or hybrid CNN-Transformer models. These models have shown promising results in medical image analysis and could improve the accuracy and efficiency of retinal disease classification.

Another possible enhancement is the expansion of the system to detect multiple retinal diseases. Future versions of the system could be trained to identify other eye conditions such as glaucoma, macular degeneration, and retinal detachment. This would allow the system to function as a comprehensive retinal disease diagnostic platform.

The system could also be integrated with mobile healthcare applications and cloud-based platforms. This would allow doctors and healthcare workers to upload retinal images from remote locations and receive automated diagnostic results in real time. Such solutions could significantly improve access to eye care services in rural and underserved regions.

Another future improvement involves integrating explainable artificial intelligence (XAI) techniques into the system. These techniques can provide visual explanations of the deep learning model's predictions, helping doctors understand which regions of the retinal image influenced the diagnosis.

Furthermore, the system could be connected with electronic health record (EHR) systems to automatically store patient diagnostic results and support long-term patient monitoring.

9.4 Applications

The Fake News Detection and Visualization Platform has several practical applications in different domains.

One important application is in journalism and media organizations, where journalists can use the system to verify the credibility of news articles before publishing them. This can help maintain journalistic integrity and prevent the spread of misinformation.

The system can also be used by social media platforms to identify misleading content shared by users. Automated fake news detection tools can help reduce the spread of misinformation across social networks.

Another application is in government and public policy organizations, where authorities can monitor and identify fake news that may cause social unrest or misinformation during important events such as elections or public health crises.

Educational institutions can also benefit from this system by using it as a learning tool for media literacy, helping students understand how fake news spreads and how it can be detected.

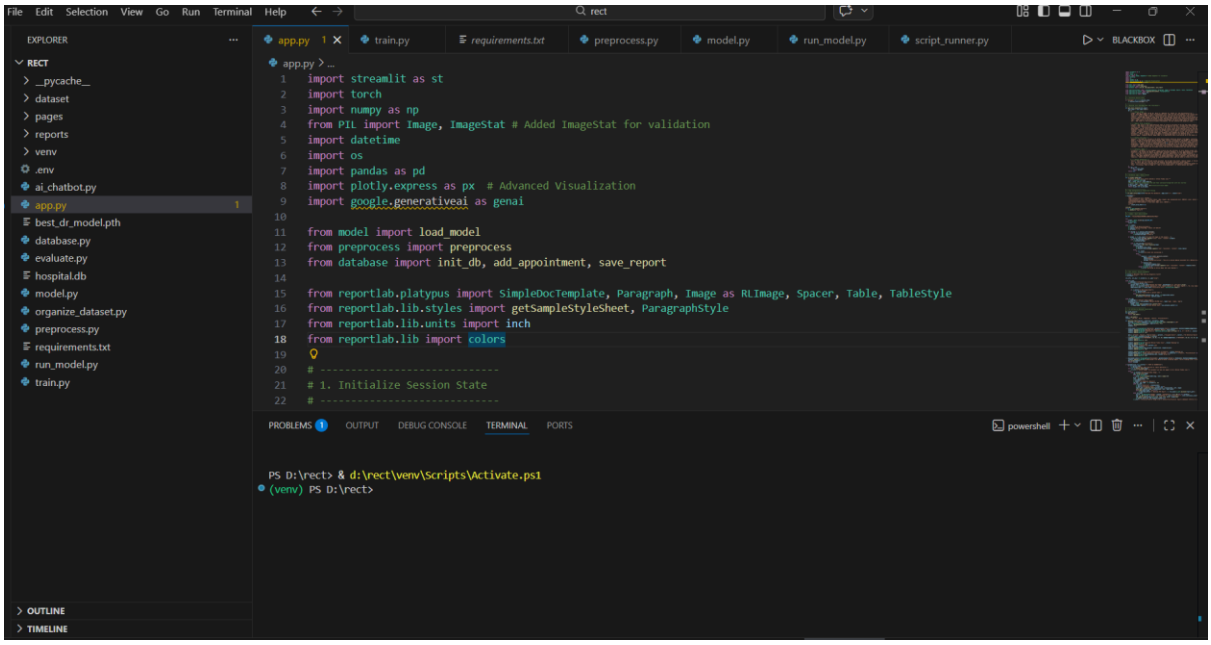
Finally, the platform can be used by general internet users who want to verify the authenticity of news articles before sharing them online.

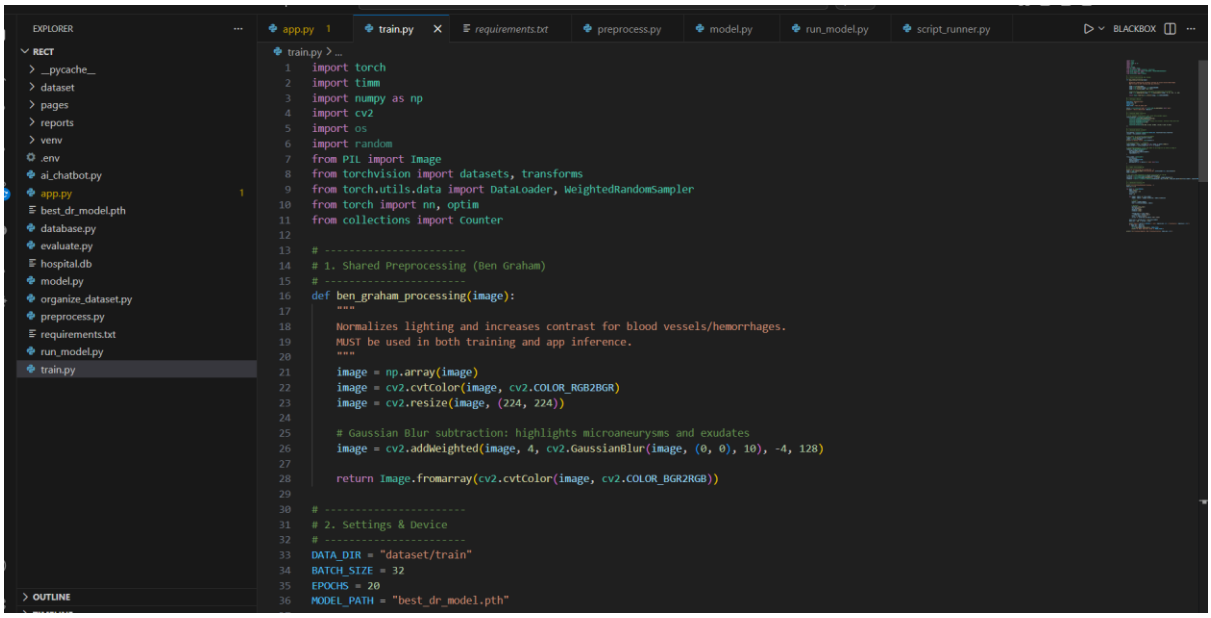
CHAPTER 10

APPENDICES

Click the below link to open the project

https://retinaaibyarj.streamlit.app



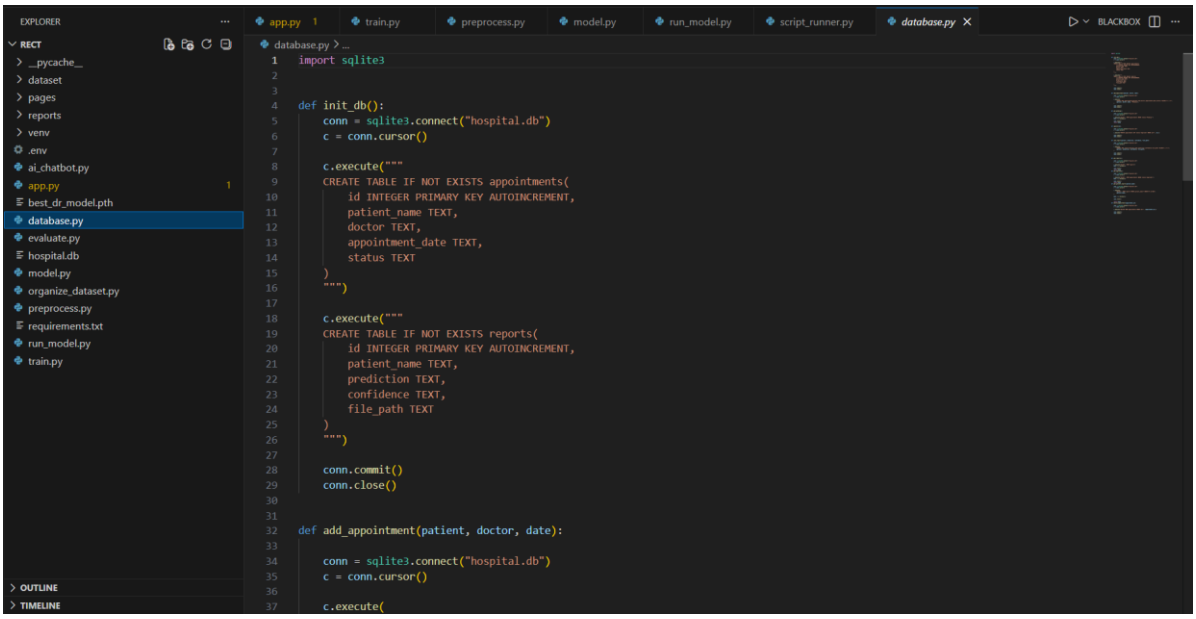


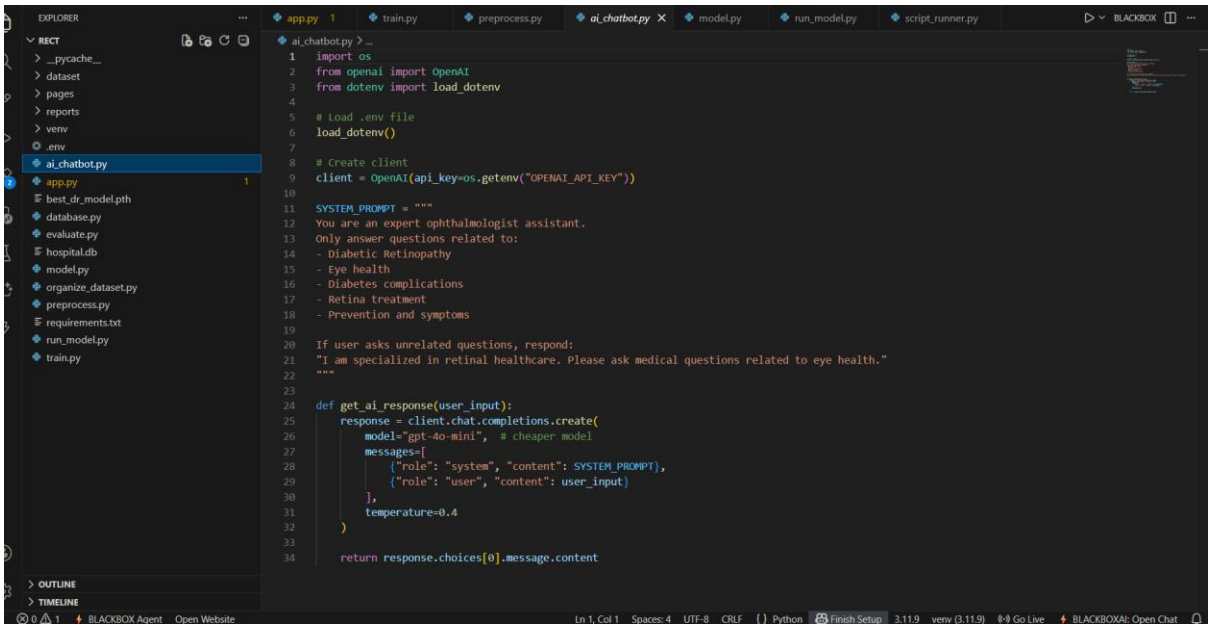
```python
import torch
from torchvision import transforms
from PIL import Image
import cv2
import numpy as np

def ben_graham_processing(image):
    image = np.array(image)
    image = cv2.cvtColor(image, cv2.COLOR_RGB2BGR)
    image = cv2.resize(image, (224, 224))
    image = cv2.addWeighted(image, 4, cv2.GaussianBlur(image, (0, 0), 10), -4, 128)
    return Image.fromarray(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))

def preprocess(image):
    # This transform must be identical to the one in train.py (excluding augmentations)
    transform = transforms.Compose([
        transforms.Lambda(ben_graham_processing),
        transforms.ToTensor(),
        transforms.Normalize([0.485, 0.456, 0.406], [0.229, 0.224, 0.225])
    ])
    # Add batch dimension: [1, 3, 224, 224]
    return transform(image).unsqueeze(0)
```

```python
import torch
import timm
import os

def load_model(model_path="best_dr_model.pth"):
    """
    Loads the EfficientNet-B0 model weights for inference.
    Ensures the model is mapped to CPU for deployment compatibility.
    """
    # 1. Recreate the exact architecture used in training
    # We set pretrained=False here because we are loading our own local weights
    model = timm.create_model('efficientnet_b0', pretrained=False, num_classes=5)

    # 2. Check if the weight file exists
    if not os.path.exists(model_path):
        raise FileNotFoundError(f"Model weight file '{model_path}' not found. Please run train.py first.")

    # 3. Load the state dictionary
    # map_location=torch.device('cpu') ensures it works on computers without GPUs (like Streamlit Cloud)
    state_dict = torch.load(model_path, map_location=torch.device('cpu'))
    model.load_state_dict(state_dict)

    # 4. Set to evaluation mode
    # This is CRITICAL: it disables Dropout and BatchNorm updates during prediction
    model.eval()

    return model
```

```python
import torch
import timm
from PIL import Image
from torchvision import transforms

classes = [
    "No DR",
    "Mild",
    "Moderate",
    "Severe",
    "Proliferative DR"
]

device = "cuda" if torch.cuda.is_available() else "cpu"

model = timm.create_model("efficientnet_b0", pretrained=False, num_classes=5)

model.load_state_dict(torch.load("best_dr_model.pth", map_location=device))
model.to(device)
model.eval()

transform = transforms.Compose([
    transforms.Resize((224,224)),
    transforms.ToTensor(),
    transforms.Normalize([0.485,0.456,0.406],
                         [0.229,0.224,0.225])
])

img = Image.open("test.jpg").convert("RGB")

img = transform(img).unsqueeze(0).to(device)

with torch.no_grad():
    outputs = model(img)
    pred = torch.argmax(outputs,1).item()
```

```python
class ScriptRunner:
    def __init__(
        session_state, yield_callback=self._maybe_handle_execution_control_request
        )
        self._uploaded_file_mgr = uploaded_file_mgr
        self._script_cache = script_cache
        self._user_info = user_info
        self._fragment_storage = fragment_storage

        self._pages_manager = pages_manager
        self._requests = ScriptRequests()
        self._requests.request_rerun(initial_rerun_data)

        self.on_event = Signal(
            doc="""Emitted when a ScriptRunnerEvent occurs.

            This signal is generally emitted on the ScriptRunner's script
            thread (which is *not* the same thread that the ScriptRunner was
            created on).

            Parameters
            ----------
            sender: ScriptRunner
                The sender of the event (this ScriptRunner).

            event : ScriptRunnerEvent

            forward_msg : ForwardMsg | None
                The ForwardMsg to send to the frontend. Set only for the
                ENQUEUE_FORWARD_MSG event.

            exception : BaseException | None
                Our compile error. Set only for the
                SCRIPT_STOPPED_WITH_COMPILE_ERROR event.

            widget_states : streamlit.proto.WidgetStates_pb2.WidgetStates | None
                The ScriptRunner's final WidgetStates. Set only for the
```


EXPLORER
app.py
train.py
preprocess.py
model.py
run_model.py
script_runner.py
database.py
BLACKBOX
RECT
__pycache__
dataset
pages
reports
venv
.env
ai_chatbot.py
app.py
best_dr_model.pth
database.py
evaluate.py
hospital.db
model.py
organize_dataset.py
preprocess.py
requirements.txt
run_model.py
train.py
OUTLINE
TIMELINE
database.py > ...
import sqlite3
def init_db():
conn = sqlite3.connect("hospital.db")
c = conn.cursor()
c.execute("""
CREATE TABLE IF NOT EXISTS appointments(
id INTEGER PRIMARY KEY AUTOINCREMENT,
patient_name TEXT,
doctor TEXT,
appointment_date TEXT,
status TEXT
)
""")
c.execute("""
CREATE TABLE IF NOT EXISTS reports(
id INTEGER PRIMARY KEY AUTOINCREMENT,
patient_name TEXT,
prediction TEXT,
confidence TEXT,
file_path TEXT
)
""")
conn.commit()
conn.close()
def add_appointment(patient, doctor, date):
conn = sqlite3.connect("hospital.db")
c = conn.cursor()
c.execute(


EXPLORER
app.py
train.py
preprocess.py
ai_chatbot.py
model.py
run_model.py
script_runner.py
BLACKBOX
RECT
__pycache__
dataset
pages
reports
venv
.env
ai_chatbot.py
app.py
best_dr_model.pth
database.py
evaluate.py
hospital.db
model.py
organize_dataset.py
preprocess.py
requirements.txt
run_model.py
train.py
OUTLINE
TIMELINE
ai_chatbot.py > ...
import os
from openai import OpenAI
from dotenv import load_dotenv
# Load .env file
load_dotenv()
# Create client
client = OpenAI(api_key=os.getenv("OPENAI_API_KEY"))
SYSTEM_PROMPT = """
You are an expert ophthalmologist assistant.
Only answer questions related to:
- Diabetic Retinopathy
- Eye health
- Diabetes complications
- Retina treatment
- Prevention and symptoms
If user asks unrelated questions, respond:
"I am specialized in retinal healthcare. Please ask medical questions related to eye health."
"""
def get_ai_response(user_input):
response = client.chat.completions.create(
model="gpt-4o-mini", # cheaper model
messages=[
{"role": "system", "content": SYSTEM_PROMPT},
{"role": "user", "content": user_input}
],
temperature=0.4
)
return response.choices[0].message.content
BLACKBOX Agent
Open Website
Ln 1, Col 1
Spaces: 4
UTF-8
CRLF
Python
Finish Setup
3.11.9
venv (3.11.9)
Go Live
BLACKBOXAI: Open Chat

CHAPTER 11
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